

Seeing what's missing

Graphical and computational evaluation of imputation methodology

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Acknowledgements

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Samenvatting

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Abbreviations and symbols

n	number of units in a sample or rows in a dataset, indexed $i = 1, \dots, n$
p	number of variables in a sample or columns in a dataset, indexed $j = 1, \dots, p$
Y	partially observed sample data (sized $n \times p$), also called the (hypothetically) complete data
R	response indicator matrix (sized $n \times p$), which stores the locations of the observed ($R_{ij} = 1$) and missing ($R_{ij} = 0$) parts of Y
Y_{obs}	observed data (elements of Y_{ij} where $R_{ij} = 1$), also called the incomplete data
Y_{mis}	missing data (elements of Y_{ij} where $R_{ij} = 0$)
Q	quantity of scientific interest; the scientific estimand we would like to estimate if we had complete data
$P(Q \dots)$	complete-data model ; the analysis model we would like to fit if we had complete data, also called the ‘model of scientific interest’
$P(R \dots)$	missing data model ; the model that relates Y to R , which is interpreted as the ‘probability to be missing’; also called the ‘response model’
X	fully observed sample data for covariate(s) of Y , used in missing data models
ψ	parameter(s) of the missing data model that represent any cause of the missingness unrelated to X and Y
MCAR	‘Missing Completely At Random’; missing data mechanism with missing data model $P(R \psi)$, also called ‘not data dependent’
MAR	‘Missing At Random’; missing data mechanism with missing data model $P(R Y_{\text{obs}}, X, \psi)$, also called ‘seen data dependent’
MNAR	‘Missing Not At Random’; missing data mechanism with missing data model $P(R Y_{\text{obs}}, Y_{\text{mis}}, X, \psi)$, also called ‘unseen data dependent’
LWD	list-wise deletion
CCA	complete case analysis
MI	multiple imputation
JM	joint modeling; imputation technique
FCS	fully conditional specification; imputation technique
SMC-FCS	substantive model compatible fully conditional specification; imputation technique
MICE	multivariate imputation by chained equations algorithm
mice	multivariate imputation by chained equations software
m	number of imputations, indexed $\ell = 1, \dots, m$
$P(Y, X, R)$	joint distribution of the incomplete data, fully observed covariate(s), and the missingness

$P(Y_{\text{mis}} \dots)$	joint (imputation) model, where $P(Y_{\text{mis}} Y_{\text{obs}}, R)$ denotes the posterior distribution of the missing data conditional on the observed data and the missingness
Y_j	univariate subset of Y (sized $n \times 1$); an incomplete variable
Y_{-j}	all data in Y except for column j (sized $n \times (p - 1)$)
$P(Y_{\text{mis},j} \dots)$	imputation model for column j , where $P(Y_{\text{mis},j} Y_{\text{obs},j}, Y_{-j}, R)$ would indicate all other variables are imputation model predictors
$\dot{Y}_\ell$	imputed data in imputation number ℓ (of m total)
Y_ℓ	completed data in imputation number ℓ , consisting of $\{Y_{\text{obs}}, \dot{Y}_\ell\}$
$\hat{Q}_\ell$	estimate of Q calculated from completed data in imputation number ℓ
$\bar{Q}$	pooled estimate based on m imputations
T	total variance of $Q - \bar{Q}$
U	within-imputation variance due to sampling
B	between-imputation variance due to nonresponse
SE	standard error
CI	confidence interval
γ	fraction of information missing due to nonresponse, also called ‘fraction of missing information’ (FMI)
PMM	predictive mean matching; imputation method
V&V	verification and validation of (software) systems, where verification refers to the process of evaluating a system’s output, and validation assesses its fit for purpose.

1 Introduction

TL;DR (or: the quality of the solution)

This dissertation is about algorithms and the quality of their output. I investigate how one specific type of data-generating algorithms can be evaluated for use in statistical inferences. The type of algorithms in question are imputation algorithms: algorithms with the purpose of generating plausible data values, based on a model of some observed information. This type of algorithm is popular for solving missing data problems.

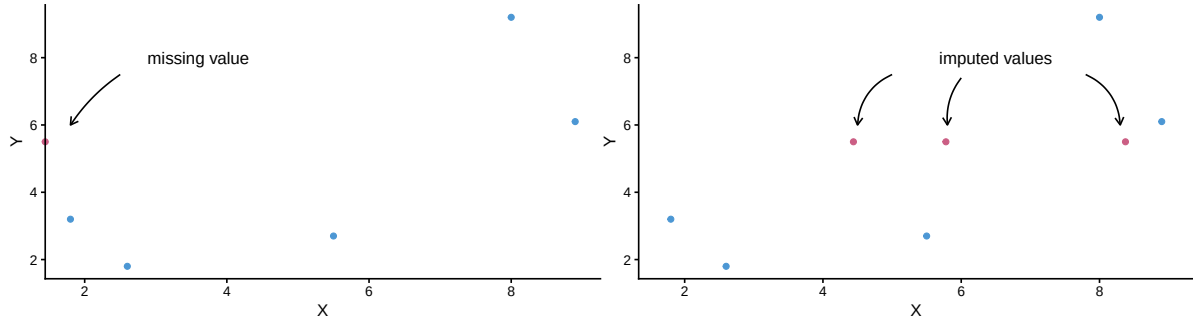
Missing data problems are timeless, but technological advances have made treating missingness easier than ever. A single line of code can seem to solve your missing data problem nowadays. But is that solution really what you expect? That is what my dissertation is about. I want to contribute methodology that facilitates drawing valid inferences from incomplete data.

Missing data are ubiquitous and cannot be ignored without risking bias. Missing data occur often in scientific disciplines with human subjects, such as the social sciences and biomedical research. A participant in a study might, for example, drop out of a longitudinal trial or refuse to answer a sensitive question, resulting in gaps in the research data. These gaps complicate data analysis and may lead to biased and invalid study results. Imputation algorithms are designed to ‘impute’ (i.e., fill in) missing values in incomplete datasets. Imputation is a popular and flexible procedure that separates the missing data problem from the analysis of scientific interest.

Imputation allows data analysts to first solve for the missingness, to then be able to analyze the completed data as if they were completely observed. If done several times in parallel, imputation can yield estimates that are both unbiased as well as confidence valid. The practice of drawing inference by replacing missing values with several synthetic substitutes is called ‘multiple imputation’ (MI). Figure 1 provides an example of multiple imputation, where a missing value is replaced several times with plausible substitutes. The variability between imputations will reflect the uncertainty due to non-response in the final estimate.

Although imputation is widely supported by software and many implementations of the method have convenient defaults, generating good imputations is not a trivial task. **How can we know when to trust the output of imputation algorithms?** There are three approaches. The ideal route is to study the formal statistical properties of the method via analytical derivations. Unfortunately, this is often impossible for imputation algorithms, many of which do not have formal proofs. Instead, the de facto standard is to estimate their empirical performance via simulation studies. These studies, however, are not standardized and are therefore hard to compare. Moreover, simulation study results might not always translate to applied use. Thirdly, for these real-world applications there is no systematic framework for evaluating the practical performance of imputation algorithms, such as diagnostics for algorithmic convergence. Ergo, evaluating whether imputations should be trusted remains challenging.

Figure 1: Multiple imputation of a missing value



In this dissertation, I set out to investigate what is needed to develop, apply, and evaluate data-generating algorithms. I thereby focus on the imputation procedure ‘MICE’ (van Buuren & Groothuis-Oudshoorn, 1999) as implemented in the R package `mice` (van Buuren & Groothuis-Oudshoorn, 2011).

Why? Because current evaluation practices (simulation design, convergence diagnostics, visual diagnostics) are fragmented and insufficient.

How? By bridging inferential evaluation and data-/algorithm-level diagnostics for imputation methods.

1.1 Background

Why this dissertation? (or: certainty with uncertainty)

My academic tagline has long been “statistician, interdisciplinary, open scientist”. And while there’s no doubt I am an interdisciplinary and an open scientist, I am no longer keen on calling myself a statistician necessarily. I am more inclined to say I am a methodologist rather than a statistician. As a methodologist, I am interested in how science is done, and how it can be done better. Especially in the social and biomedical sciences, research should not be purely quantitative. Statistics is just *one* way of doing science.

Don’t get me wrong, statistics is still a favorite of mine. I love stats (I even have a mug that says so). Statistical inference allows us to distinguish between real-world effects and random noise (Rosenberg, 2018). Epistemologically speaking, statistical inference may be classified either as deductive or inductive inference; there is no consensus in the literature about which pattern of scientific reasoning suits statistical inference (de Ruiter & Albert, 2017). Gelman & Shalizi (2013) argue that statistical inference is deductive in the context of hypothesis testing (i.e., when general laws from mathematical probability theory are used) and inductive in the context of parameter estimation (i.e., when sample statistics are extrapolated to the population). In

other words, statistics is the science of uncertainty and extracting information from data (Hand, 2025). The latter can feel magical at times: the process of turning data into insights. But I am most a fan of the uncertainty aspect.

I am motivated to contribute to trust in science by making it easier to reflect uncertainty in data analysis efforts, inferences, and graphs. In this dissertation, I hope to achieve that for missing data methodology.

But first, some terminology and notation

Statistical analyses are a means to infer conclusions about a population after observing only a subset of this population: a sample. If the sample is representative, sample estimates will reflect the population characteristics plus some uncertainty surrounding it. Statisticians call this uncertainty ‘errors’ (e.g., the ‘sampling error’ due to observing only a subset of the population), which is often expressed in terms of the standard error (SE) of an estimate. Standard errors can get translated into decision tools for inference such as p -values and confidence intervals (CIs), which help researchers distinguish between signal and noise. CIs, for example, should cover the true population parameter at the frequency of the nominal confidence level (e.g., 95%).

A statistical procedure with a coverage probability that matches the nominal confidence level is called ‘confidence valid’ (Neyman, 1934). Confidence validity provides an operational link between the mathematics of probability and empirical practice. Confidence-valid statistical methods give us certainty about the uncertainty in our inferences from a sample to the population. So, statistics is a means of estimating effects within a sample and inferring back to the population with some expression of uncertainty.

Uncertainty in statistical inferences can stem from many sources. In the ‘Total Survey Error’ framework (Deming, 1944), other sources of uncertainty are described such as non-response error. Non-response error occurs when participants do not (fully) respond to data collection efforts. We speak of unit non-response for intended study participants who do not provide any data at all (for example because they could not be reached, or refused to participate). Item non-response refers to patterns of missingness where some—but not all—data are missing for a participant (for example because they skipped a survey question). Item non-response is the focus of this dissertation: the way we handle incomplete observations in partially observed data and how that affects our inferences.

I will not pretend to provide an encompassing overview on treating missing data problems. Others have written much more extensive reference works on the matter, see for example Rubin (1996), Schafer (1997), Allison (2002), Schafer & Graham (2002), Graham (2012), Molenberghs et al. (2014), van Buuren (2018), and R. J. A. Little & Rubin (2019).

How we treat non-response is reflected in the uncertainty of our inferences. Unit non-response is often treated using weighting, which may introduce ‘weighting error’ Alsalti et al. (2026). It

is not typically part of the TSE framework, but item non-response treatment can also produce error, such as ‘imputation error’ (Federal Committee on Statistical Methodology, 2001).

Non-response bias can be minimized by preventing missingness altogether: the best way to treat missing data is by not having any (a truism attributed to Gertrude Mary Cox). Whenever missing data cannot be prevented, the only way forward is to treat the missing data problem. Missing values force data analysts to treat the missingness in some way.

Why are missing data problematic?

Let’s use an empirical example. Generally speaking, statistical inference works really well. For example, when sociologists wanted to study trust in the government during the COVID-19 pandemic, it was not feasible to survey all Dutch citizens. Instead, they used a representative sample. Some thousands of people were asked to fill in a questionnaire about their trust in the government over the course of the pandemic. By repeating the questionnaire over time, the researchers wanted to gain insight into the development of (dis)trust among different subgroups in Dutch society. The estimates from this sample would have been used to infer back to the population and inform policy, if it had not been for missing data...

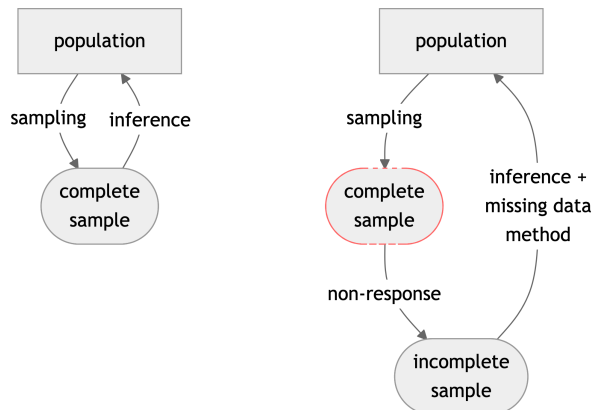
Missing data are a nuisance because it is impossible to do statistical inferences with incomplete data. When there is just one missing value in a dataset, statistical inference cannot be performed. With incomplete data, even simple statistics such as the mean of variables are mathematically undefined. To obtain results from your incomplete dataset, you will always employ a missing data strategy—whether deliberately or not.

To get an estimate from incomplete data, some kind of missing data method has to be employed. Even without specifying a missing data method explicitly, there will be handling of the missing data. The default strategy in most statistical software packages is to ignore all cases that have missing values (van Buuren, 2018). This is referred to as ‘list-wise deletion’ (LWD). Ignoring missingness, however, lowers statistical power and may yield invalid inferences (e.g., biased estimates, spurious significant results, see e.g., van Buuren, 2018). List-wise deletion is ill-advised under most conditions (for exceptions see e.g., van Buuren, 2018 § 2.7; Carpenter & Smuk, 2021, § 3.1). List-wise deletion results in bias because it assumes that the observed part of the data is equivalent to the missing part of the data, and this is not usually the case.

Why are the missing and observed parts not equivalent? (or: the missing data mechanisms)

List-wise deletion (complete-case analysis) assumes that the observed part of the data is representative of the missing part of the data, which seldom holds in practice. In practice, the missing part of the data is often missing due to some non-trivial reason, other than pure chance (or ‘bad luck’). If there is a non-chance cause for the missingness across observations, the observed part of the data is not representative of the missing part of the data. This relationship

Figure 2: Inference with incomplete data



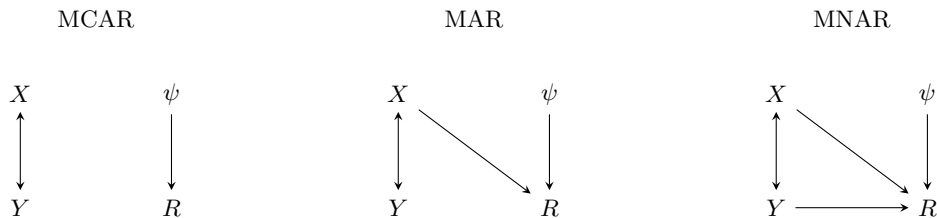
Note. Schematic view of statistical inference in the presence of non-response. Left: no missing data in the sample from the population (i.e., ideal circumstances/business as usual). Right: incomplete sample due to non-response, so no complete sample could be obtained. What types of missing data problems are we considering? We are interested in data with incomplete cases and/or variables, or 'item non-response'. Incomplete samples where some—but not all—entries are missing per row or per column. Entirely missing rows would be 'unit non-response'; entirely missing columns would be unobserved variables.

can be formulated in terms of a missing data model. Missing data models capture the causal relation between the data (Y) and the associated missingness indicator (R). Missing data models thereby reflect the 'probability to be missing' for any given observation. Missing data models are used to describe and classify the causes of the missingness, which determines how the missingness should be handled to arrive at unbiased and confidence-valid results.

There are three missing data mechanisms described by missing data models. If there is no data-related cause of the missingness, the missingness is said to be 'completely at random' (MCAR), or 'not data dependent' (respective terminology by Rubin, 1976; and Hand, 2020). If there is an association between the missingness and the observed part of the data, the missingness is 'at random' (MAR; conditional on the observed data) or 'seen data dependent'. If the cause of the missingness is missing too, the missingness is 'not at random' (MNAR), or 'unseen data dependent'. Figure 3 provides a graphical representation of these three mechanisms, where the missingness (R) in incomplete data Y either depends on Y itself (i.e., MNAR), on observed covariate(s) X (i.e., MAR), or none of these (i.e., MCAR).

Modeling the non-response can aid the analysis of incomplete data. Together with complete-data models (i.e., the model of scientific interest), data models inform data analysts how the missingness might affect the scientific estimates. For example, in that study about trust in the government during the COVID-19 pandemic, it was not random who had missing data on the question about trust in the government. People with lower trust in institutions were also less likely to respond to the questionnaire. This resulted in higher missingness rates across

Figure 3: Missing data mechanisms



Note. Incomplete data Y and fully observed covariate X covary. Under missing completely at random missingness (MCAR), the missingness indicator R is only affected by ψ , causes of the missingness unrelated to X and Y . Under MAR, the missingness also depends on observed information (X), whereas MNAR entails missingness that is dependent on the missing data itself (Y). Figure adapted from Schafer & Graham.

participants with lower trust in the government. Using only the observed data to infer to the population would then leads to biased estimates (over-estimation) of trust in the government. Methods that implicitly assume MCAR (like list-wise deletion) are therefore likely to cause bias.

In human subjects research, pure MCAR missingness is unlikely. Only truly random missingness, independent of any individual characteristics, would amount to missing completely at random missingness mechanism. This could happen, for example, due to a glitch in the service software, affecting all participant equally likely. In our case, however, there were clear signs of non-MCAR missingness. If we can model the missingness probability based on known data, such as previous observations or demographic data, we have ‘Missing At Random’. Then, conditional on this information, observed and missing values have the same probability to be missing. Under MAR, we can model the missingness probability using known variables and obtain valid inference if this model is adequate. This is in contrast to the ‘unseen data dependent’ missingness mechanism or ‘missing not at random’. If the missingness depended the unobserved values themselves, we speak of ‘Missing Not At Random’. For example, when people who distrust the government are less likely to respond to the trust item, and we have no way of correcting for this. This kind of missingness cannot be resolved from the observed data alone and requires assumptions or external information. In practice, it is impossible to know from the observed part of the data alone whether we are dealing with MNAR, so you always have to make an assumption about the mechanism underlying the missingness in your data.

If the missingness is MNAR, the validity of the inferences is at risk. MNAR missingness is often called ‘non-ignorable’, a term which roughly means that given the observed data, you cannot treat the missing-data mechanism as irrelevant for the subsequent inferences. This is in contrast to ‘ignorable’ missing-data mechanisms, which meet two requirements: 1) conditional on the observed data, the probability that an entry is missing does not depend on the unobserved value itself (i.e., MCAR or MAR missingness), and 2) the parameters in the missing data model are distinct from the parameters of the substantive model (Little & Rubin, 2020). If treated correctly, ignorable missingness will yield valid inferences.

How to treat the missing data problem? (or: ignore, model, fill in – the missing data methods)

This section describes three popular routes for treating missing data. I will not pretend to provide an encompassing overview on treating missing data problems. Others have written much more extensive reference works on the matter, see for example Rubin (1996), Schafer (1997), Allison (2002), Schafer & Graham (2002), Graham (2012), Molenberghs et al. (2014), van Buuren (2018), and R. J. A. Little & Rubin (2019). The three missing data treatments I will describe are deletion-based methods, likelihood-based methods, and imputation.

The first missing data treatments are deletion-based methods, such as list-wise deletion. With list-wise deletion, we only analyze rows in the data without missing values (i.e., complete cases). Deletion-based methods are easy to apply, and sometimes even applied by default in statistical analysis software. If the missingness level is low, deletion-based methods do little harm, and under some conditions they might even be the best missing data method to apply (see e.g., van Buuren, 2018 § 2.7; Carpenter & Smuk, 2021, § 3.1). Deletion-based methods, however, do not generally accommodate differences between the observed and missing parts of the data, making them mostly applicable to MCAR. Under most kinds of missingness, deletion-based methods will therefore risk biasing the statistical inferences. Moreover, these methods discard incomplete observations, resulting in lower power and potential resource waste compared to non-deletion-based methods. In short, deletion-based methods do not treat the missing data as much as ignore them, and one should be careful of bias when applying them.

Secondly, there are methods that incorporate the missingness into the model of scientific interest. Modeling methods do accommodate MAR, but are often quite complex. These missing data methods jointly specify the complete-data model and missing data model per analysis. For example, Bayesian methods can incorporate the missingness into the analysis model. But you have to specify the model for each analysis separately, and there are no null hypothesis significance tests which is a problem when working with applied researchers. Another option to incorporate the missingness in the analysis model is expectation maximization. But this yields many parameters and might not be available for a complex multilevel model like the COVID-19 data. Both of these techniques are tied to a specific analysis model, which might not be suitable for contemporary practitioners who run several models in a data science workflow.

Thirdly, there are imputation methods. Imputation methods do not have these drawbacks of being computationally advanced and analysis-model specific. Imputation splits the missing data problem from the analysis of scientific interest. Imputation methods first solve for the missingness, to then analyze the completed data ‘as if’ they were fully observed. This allows the use of familiar complete-data methods instead of bespoke incomplete-data models. Imputation is useful when different people are responsible for different parts of a project (e.g., a missing data methodologist vs. domain expert). And when many different analyses are planned on the same data, it is possible to run several analyses without having to incorporate the missingness into the scientific model each time. This makes imputation better suited for contemporary data science workflows than missing data methods that incorporate the missingness in the

substantive model. Compared to missing data methods that incorporate the missing data like Bayesian methods or likelihood-based approaches (e.g., EM), imputation is more flexible and often less complex for non-experts. Compared to list-wise deletion, imputation methods offer the ability to accommodate MAR mechanisms, and they reduce research waste by retaining incomplete cases. Imputation thus has the ability to produce unbiased estimates. But are they properly uncertain (i.e., confidence-valid) too? That’s where multiple imputation (MI) comes in!

Why multiply impute? (or: several ‘wrongs’ *can* make a ‘right’)

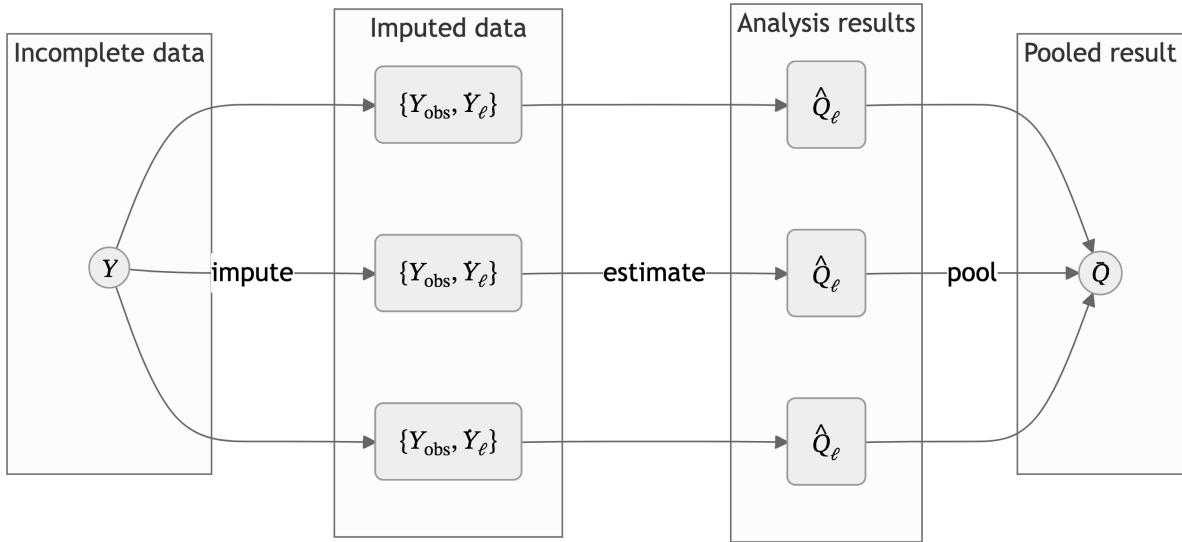
All techniques that rely on a single replacement value risk under-representing uncertainty in the inferences, because the uncertainty due to the non-response is not taken into account. The uncertainty due to non-response can be captured using the multiple imputation framework. Conceptualized by Rubin (1976), MI posits that repeated stochastic draws per per missing datapoint create a distribution of plausible values: the posterior distribution of the missing data conditional on the observed data and the missingness, $P(Y_{\text{mis}}|Y_{\text{obs}}, R)$. This posterior distribution reflects the uncertainty due to non-response. By drawing imputations from the posterior distribution of the missing data, we obtain completed versions of the data that *might have* been.

MI is the practice of generating several completed datasets by replacing each missing value with a synthetic substitute multiple times. The completed datasets can be analyzed as if the data were complete to begin with. On each completed dataset, we obtain estimates of the analysis of scientific interest. These estimates per imputation are then combined to obtain one pooled estimate. The variance of the estimate incorporates not just the sampling variance (within each imputation) but also added variance due to non-response (between imputation variance and simulation error). The variability between imputations reflects uncertainty due to the missingness.

The result is a probability distribution for each missing value, from which we subsequently draw several values as imputations. If we would use infinitely many values as imputations, we would obtain the entire probability distribution. In practice, however, we use a limited number of imputations. With that, we approximate the probability distributions of missing values. The more imputations are used, the better we can represent the uncertainty in the data due to missingness. This is expressed in the ‘simulation error’.

Multiple imputation means that each missing value is filled in several times, with slightly different model parameters to represent model uncertainty. The result of MI is that the uncertainty in the data due to missingness is preserved in the inference. Correcting for missingness happens at two levels: 1) modeling the relation between missing cases and covariates; 2) adding the appropriate variance to the models to represent the uncertainty due to missingness. Under the right conditions, MI yields valid, unbiased estimates (Schafer, 1997). MI enables researchers with missing data problems to correct for differences due to missingness and maintain the

Figure 4: Multiple imputation scheme ($m = 3$)



Note. Read left to right. Incomplete data Y is partially observed, $Y = \{Y_{\text{obs}}, Y_{\text{mis}}\}$. The missing data Y_{mis} is imputed m times to obtain m completed datasets $\{Y_{\text{obs}}, \tilde{Y}_{\ell}\}$, where $\ell = 1, 2, \dots, m$. On each imputation, the quantity of scientific interest Q is estimated, yielding m analysis results $\hat{Q}_{\ell}$. These results are pooled across imputations to obtain a single pooled result $\bar{Q}$.

appropriate level of uncertainty. The aim is to generate imputations to fill in the missing part of the data. How do we model the missing part of the data based on the observed part of the data?

1.2 Imputation in practice

How to generate imputations? (or: univariate, joint, chained – the imputation flavors)

We model the distribution of this missing part based on what is observed, using the observed part of the data to fix the missing part of the data. Which types of model or models do we use?

The simplest model is no model at all. With hot deck imputation, we use random other person's observation to impute or fill in the values for people with a missing data point. The problem is that this disrupts the multivariate distribution of the data, so we get biased estimates. Another simple technique is to impute a univariate statistic such as the mean. Unfortunately, imputing the mean will bias your scientific estimates even worse than list-wise deletion would, because mean imputation severely biases any multivariate statistics downward, and there is no correction for the missing at random missingness mechanisms. In general, all univariate imputation methods fail to retain multivariate associations in the data, and therefore cannot account for MAR mechanisms. So, instead of imputing variables univariately, imputations should retain the multivariate structure of the data.

Capturing the multivariate structure of incomplete data is historically done in two steps: a modeling step and an imputation step. In the modeling step, a multivariate distribution of the full dataset would be defined, after which imputations could be drawn from the joint model. In a joint model, one specifies $P(Y, X, R)$: the distribution of the incomplete data, complete covariates, and missingness indicator. Joint modeling imputation was originally developed for known distributions, such as multivariate normal data. With increasing data complexity and dataset size, specifying a joint model might not be feasible. The problem with joint modeling is that there might not exist a multivariate probability distribution, for example, with factor variables involved. Joint modeling is a direct and inflexible way to arrive at a joint distribution. Good thing we do not need this joint model explicitly.

Missing observations can be filled in using one or more imputation models instead. While Rubin (1987, pp. 160–166) “distinguished three tasks for creating imputations under an explicit model: the *modeling* task the *imputation* task and the *estimation* task.” (van Buuren, 2007, p. 221; in Molenberghs et al., 2014), FCS does not require these explicit tasks. In the FCS framework (van Buuren et al., 2006), we only need an imputation model for each incomplete variable. Imputation models consist of the functional form of the model (e.g., stochastic regression) and imputation model predictors (i.e., other variables in the data). FCS is substantive model agnostic, and does not assume any specific joint distribution. Instead, FCS only requires an *imputation model* to describe how synthetic values may be generated, without explicit

specification of the joint model. This makes FCS flexible. Just specify an imputation model per variable.

How to define imputation models?

We define an imputation model per incomplete variable. And we need two things for this: 1) We need to specify the functional form of the model. For example a semi parametric method such as predictive mean matching for Likert scale items such as trust in the government. And 2) we need to choose imputation model predictors. For example, variables with a high linear association with this incomplete variable, or variables that are related to the missingness indicator of this incomplete variable (such as demographic data).

Some things to keep in mind: With FCS, the quality of the imputation model can only be determined/assessed based on the complete data model. But what if that model is not (yet) available? Then take the most complex possible model and check that, because if you would take the simplest model, then that would implicitly mean you are constraining parameters (forcing $b=0$), systematically suppressing relations. Don't forget to add the effect of interest in your model of scientific interest. Otherwise, you are constraining the relation between the variables to be zero. If there is a specific data analysis model for the effect of scientific interest that you already have in mind, consider substantive model compatible fully conditional specification. SMC-FCS (Bartlett et al., 2015) is optimized for known analysis model models: we assume a complete data model and derive the imputations from there. But what if the analysis model is unknown? Well, Fully conditional specification is the substantive model agnostic. So we build an imputation model based on the most complex analysis model that we might want to test.

FCS implicitly relies on the joint distribution. The nice thing about joint modeling is that convergence of the model is guaranteed, whereas this is not the case with fully conditional specification. "Imputation models bypass the need to specify $P(Y, X, R)$, though their use creates new responsibilities for substantiating its correctness for a given statistical analysis." (van Buuren, 2007, p. 222). The question becomes: how do we choose an appropriate imputation model? And how do we know whether we have chosen an appropriate imputation model?

FCS has no deductive or analytical proof, so we have to rely on evidence from simulation studies. It has been shown that FCS works under many conditions. But how to know it works in practice? We need assumptions. Many modeling assumptions inherent to a model-based technique like MI cannot be tested without the missing data itself. So, like assumptions for GLM, we look at the data to evaluate the appropriateness of our chosen model. Similarly, we need to evaluate the choice of imputation models for signs of assumption violations. Some assumption checks are visual inspection, some are quantitative, some need external info.

How to evaluate imputation models?

How do methodologists know whether an imputation method works? Because we have empirical estimates from simulation studies (and, if available, formal properties from analytical derivations). How do imputers know whether their imputation models worked? Our current answer is well, they don't know for sure. And that's what this dissertation is all about.

Life cycle of imputation methods: method is theorized, method is implemented in software, method is evaluated in simulation study, method is put to scrutiny by scientific community, method is applied in practice, method is evaluated in real-world data.

There are three main routes to evaluate imputation models. One can evaluate:

- the formal statistical properties of the method via analytical derivations;
- the empirical performance (estimates of the method's attributes) via simulation studies; and
- the operational utility (practical usefulness) in real data.

The first and preferred option is to obtain the formal statistical properties of the imputation method using analytic derivations. Often this is impossible or unfeasible with fully conditional specification, when no known joint model exists. Without this, we cannot definitively conclude that an imputation method is an unbiased estimator. Instead, we estimate the empirical performance of these methods using simulation studies. This is the default evaluation in the missing data methodology field.

Simulation studies are the standard, but not standardized. If we have a specific analysis model of interest, we can calculate performance metrics such as bias and confidence interval coverage to conclude whether a method is unbiased and confidence valid for a certain estimand. This requires a specific analysis model, which might not be the case. And it is impossible to cover all the potential use cases. Another problem is that this is not standardized, so it's difficult to compare different simulation studies. We could also calculate the imputation accuracy by comparing the imputations against the complete data, but this is ill-advised.

The third option is the operational utility. This is case-by-case evaluation in applied use with real data. In practice, confidence validity is not guaranteed. Multiple imputation aims to propagate uncertainty due to missingness, but only under specific conditions such as correct assumptions about the missingness mechanism, 'congeniality' between imputation and analysis models, and adequate algorithmic behavior. When these conditions fail (e.g. under MNAR missingness, incomplete imputation models, or non-convergence in the algorithm), the estimates may not be unbiased and/or confidence-valid. Therefore, we check the imputation model itself, for example to see whether the algorithm has converged. We also inspect the output and we might run sensitivity analyses. This is not systematic at all currently. That is a problem. Users of the method then do not know whether they have correctly applied the method.

In this dissertation, I focus on the second and third routes. We will borrow some terms from related fields. I will connect it to concepts from systems and software engineering, and the information retrieval sub-field of computer science.

1.3 Evaluation tools

Imputation as a system

From a computer science perspective, imputation workflows are ‘sociotechnical systems’. In contrast to a purely technical system, a sociotechnical system is a collection of interrelated components that delivers services to users, consisting of software, hardware, *and* human operators (Sommerville, 2015, p. 556). Sociotechnical systems “include one or more technical systems but, crucially, also people” (Sommerville, 2015, p. 552), and their success relies on the joint functioning of the system components (Sommerville, 2015, p. 556). Sociotechnical systems have three characteristics, namely:

- emergent properties of the system as a whole, which only becomes apparent when integrating the separate components (e.g., dependability);
- non-deterministic outputs, resulting from human operators who may differ in the choices they make; and
- subjective success criteria, with objectives that depend on the context of evaluation (Sommerville, 2015, p. 558).

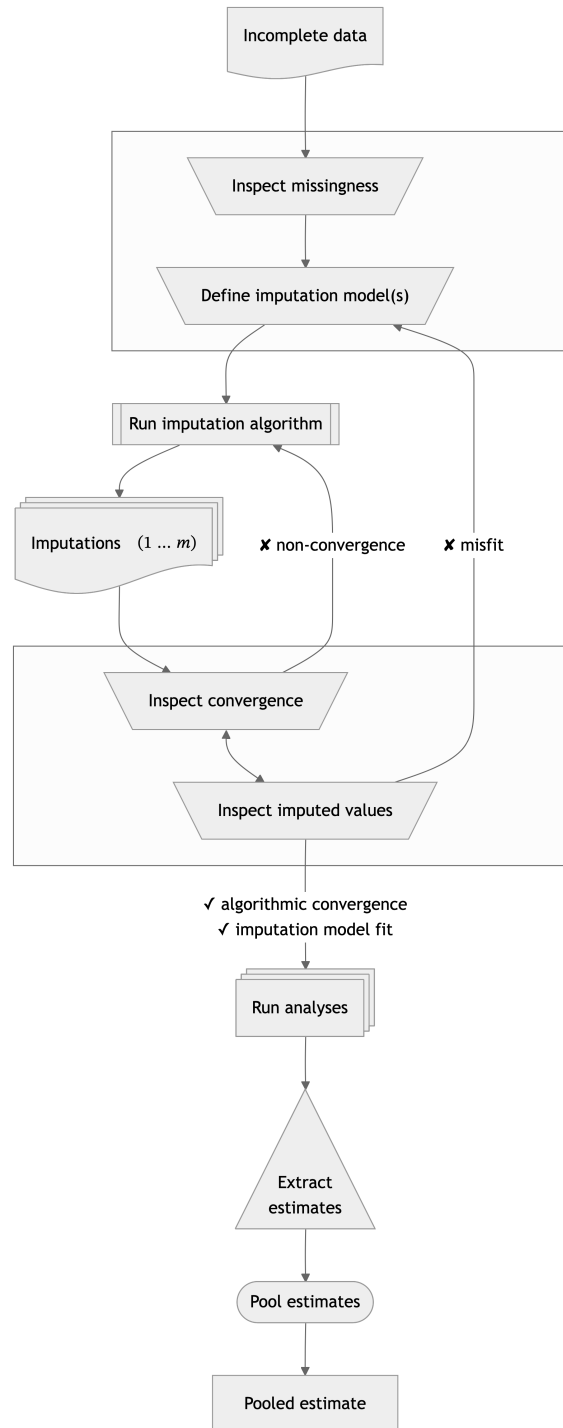
When we apply this terminology to imputation workflows, we can conclude that `mice` is a technical system: a piece of software with algorithms to generate imputations. Applications of `mice`, however, are sociotechnical. In practice, imputation workflows include `mice`, the imputation script, the analysis script, visualizations, and the researcher interpreting the output and deciding what to accept. Imputing is a sociotechnical system, which includes more components than just `mice`. Successful use of `mice` depends on the usability of the software in practice.

Building and evaluating imputation models (or: human in the loop)

Imputation with `mice` is a sociotechnical process. The software will happily produce imputations with its defaults, but *good* imputation requires human input. There are some decisions that you, as imputer, implicitly or explicitly make. Figure 5 shows where human input is requested in the imputation workflow.

A data analyst faced with incomplete data should not start imputing right away. They should first ask themselves: what kind of missingness am I dealing with? Inspect the patterns of missingness and consider which missingness mechanism(s) might be at play. A missing data pattern plot tells you where missingness occurs (which variables, which cases) and which

Figure 5: Multiple imputation flowchart



Note. Statistical evaluation is mainly focused on the relation between the incomplete data and pooled estimates. Computational evaluation is in every trapezoid: processes that require human intervention. In the first box, the question is: 'how can we model the missingness?', in the second box, the question is: 'do the imputation models fit the data?'. Missing in the graph: drawing inference with pooled estimate, and use external info after inspecting the missingness to involve all stakeholders (CRISP-DM steps).

variables tend to be missing together. For conclusions about the missing data mechanism, we need external information. Imputation with `mice` assumes ignorable missingness. MCAR and MAR are modeling assumptions, not empirically verifiable facts. In practice, it is impossible to determinately differentiate between MCAR and MNAR. But specific MAR mechanisms may become apparent through computational and graphical evaluation (e.g., by calculating or plotting associations between observed data and missingness indicators). If MNAR is suspect, use dedicated tools and/or design sensitivity analyses to assess how robust your conclusions are. If MAR mechanisms show up, include the relevant relations into the imputation models.

For each incomplete variable, `mice` asks you to specify an imputation method. This is a choice of functional form of the imputation model (e.g., logistic regression for dichotomous variables). How do we know which functional form to choose? The simplest option is to accept `mice`'s defaults, which are motivated by simulation studies and pragmatism. For example, the default method for continuous variables is the semi-parametric method `pmm`: predictive mean matching. With `pmm`, observed data entries are used as donors when imputing missing entries, conveniently yielding imputed values that lie within the range of the observed data. Diverting from these defaults might be useful in case of big, wide, or sparse data. That is, some methods can be computationally inefficient with big data (large n , large p), mathematically undefined with wide data ($n < p$), or yield suboptimal imputations because of a lack of donors in sparse data ranges. Moreover, if the column type in the incomplete data is not correctly specified, `mice` will assign incorrect imputation methods. Therefore, inspecting the distribution of each incomplete variable is advised (REF?). You can use bar charts, histograms, and density plots of the observed part of the variable to assess the distribution of the incomplete variables and determine imputation methods. Visual inspection can also be used to check whether there is skew or spikes (e.g., structural zeros) that require transformations. In short, the imputation method determines the distribution of the imputed values, which should match the distribution of the observed values to at least some extent.

The next major decision concerns the predictor matrix: which variables should be used to impute which? Some common sources of input are the associations assumed by the analysis model, associations in the observed data, associations with missingness indicators, and external input and design information. If you already know your primary analysis model (e.g., multivariate regression), it is common to include all analysis model predictors and outcomes in the imputation models. Otherwise, that would implicitly constrain these parameters to zero in the imputations, systematically suppressing relations (i.e., biasing the regression coefficients in the analysis model towards zero). In many modern workflows, the imputed data will be used for multiple, possibly unknown, analyses. You may therefore include a broader set of predictors than strictly needed for one model, or adopting 'auxiliary variables' (i.e., extra imputation model predictors that are not part of the analysis model or variables of interest). You may include predictors that are strongly associated with the incomplete variable, based on correlations ("include an auxiliary variable if its correlation with the main variable is 0.5," Nguyen et al., 2021, p. 4) Not just associations with each incomplete variable may be used to select imputation model predictors. We can also use the association of the missingness indicator with other variables in the data. Predictors that explain missingness can help you approximate MAR or correct for

design-related patterns. Other design-related patterns require expert knowledge e.g., about branching or which covariates should be included to mitigate MNAR. Choosing imputation model predictors encodes your beliefs about what information is relevant for reconstructing missing values and for capturing missingness mechanisms.

After specifying methods and predictors, you still need to ask whether the imputation model behaves adequately. We need to evaluate the imputation model post hoc. Some evaluation tools include algorithm-level diagnostics (e.g., traceplots of key parameters over iterations to assess algorithmic convergence), visual inspection of the distribution of the imputed values (e.g., to identify mismatch between observed and imputed data distributions), and sensitivity analyses (e.g., via over-imputation, which temporarily treats some observed values as missing to compare imputations to the true values).

These evaluations are neither fully automated nor purely objective. They involve building and reading plots, interpreting diagnostics, and deciding what degree of misfit is tolerable for your research aims.

Apart from the imputation models, other choices also affect the quality of the imputation workflow and resulting statistical inferences. For example, the number of imputations (m , or `m` in `mice`) and number of iterations in the imputation algorithm (`maxit`) may interact with the visual and computational diagnostics above.

Taken together, these decisions show that a `mice` workflow is not a push-button solution. It is a modeling exercise in which the user is involved in model building and evaluation. Human input makes imputation workflows subjective to individual choices and preferences.

Evaluating systems (or: verification & validation)

Since “Systems have properties that only become apparent when their components are integrated and operate together.”, we should evaluate the imputation workflow not as separate functions, but as a whole. For imputation methods this means that they should work in practice, not just in simulations. For such systems, analytic guarantees (e.g., asymptotic unbiasedness of MI) are not enough. We need computational and empirical checks on the whole workflow. And there should be interoperability with other systems an imputer might use in their imputation workflow, such as `ggplot2` for data visualization, and statistical analysis packages such as `stats`. That is where V&V comes in.

Verification and validation (V&V) is a process for the evaluation of systems, used when the ‘high road’ of formal proofs is blocked. It is framework for checking whether we have both built the system correctly as well as built the correct system.

In systems engineering, the system life cycle framework is used to build, develop and evaluate systems. There are standards for the evaluation of a system: verification & validation (IEEE Standard for System, Software, and Hardware Verification and Validation, 2025). V&V is needed when the ‘high road’ of analytical derivations and proofs are impossible or unfeasible.

Verification asks ‘was the system built right?’, whereas validation asks ‘was the right system built?’. So with verification, we can check the quality of a system by plugging in an input with a known output. Do we get the correct output in return? For imputation methodology, this is what we do with simulation studies and with unit tests in software. We want to check whether the system (or the method in this case) conforms to the requirements that we set (e.g. accuracy). Verification is about conformity to specifications, usually before or independent of real-world use.

Then validation, on the other hand, asks whether the system is suitable for operation by the actual users in a specific intended use or application. Is a system able to accomplish its intended use? Validation is use of use cases and usability testing to evaluate whether users have all they need to use the system. In stats, it’s proving that a statistical model or analytical procedure behaves as intended under real-world operating conditions. Validation is often sociotechnical: it combines technical inspection with *human judgment* and domain knowledge. A validation test is a “test to determine whether an implemented system fulfils [sic.] its specified requirements” (ISO & IEC, 2015, def. 3.4501). For imputation methodology, it is tools for users to evaluate their imputation workflow with specific data (and, optionally, their intended analysis models). This dissertation describes computational and graphical evaluation tools. In the future, we might use integration tests and user evaluation studies too?

What is computational evaluation and diagnostic visualization? Computational evaluation encompasses a range of systematic, computer-based evaluation tools such as numerical experiments or simulation studies. Computational evaluation = use of metrics, tests, and simulations to evaluate properties of the system with minimal human judgment. Diagnostic visualization is the use of graphs to communicate system behavior to stakeholders. It’s ‘diagnostic’ because the purpose is to identify patterns and problems instead of descriptive/summaries. Diagnostic visualization/visual inspection helps stakeholders understand the system’s behavior and judge whether it’s acceptable for their needs.

What is computational evaluation?

Computational evaluation is originally from information retrieval research (see e.g. “Evaluation Measures (Information Retrieval),” 2025) The practice of computational evaluation combines *effectiveness* (i.e., did the process yield the correct result, e.g., accuracy, precision, recall, etc.), *system quality* (e.g., speed, coverage of all possible results, etc.) and, importantly, *user utility* (e.g., happiness, productivity, etc.) (Manning et al., 2008) The term ‘computational evaluation’ is not specifically defined for the field of statistics, but it could be useful. One definition of computational evaluation is “to determine the quality of solutions attainable” (Dudek, 2004) Translated to statistics, a model may be statistically valid, but not fit for purpose. For example, computations that take too long to converge for use in a production pipeline/real-time use in patient care. Another definition of computational evaluation is “the process of ensuring an algorithmic solution is a good one: that it is fit for purpose” (Curzon et al., 2014). We can use this when a statistical process has different purposes and thus different usability/requirements.

For example, calculating uncertainty at the level of a single case, a sample, or inferring to a population. Since the term is not defined for statistics, there is no definition to use for the computational evaluation of imputation methodology. In this dissertation, I want to plant a flag: how can we define ‘computational evaluation’ in the context of imputation methodology? Working definition: *systematic assessment of the algorithmic behavior and data-level outputs of imputation procedures*, focused on: algorithmic stability (convergence, failure modes), sensitivity to modeling choices, plausibility and coherence of imputed values, and usability for applied researchers. These aspects are often already taken into account by imputers, but I try to systematize and operationalize these aspects that are often informal or *ad hoc*.

Computational evaluation is not equal to statistical evaluation. Computational evaluation goes beyond statistical evaluation. Statistical evaluation is important, e.g. to quantify bias and variance of estimators/procedures, but not the focus of this dissertation. Computational evaluation is something to check *conditional on* acceptable statistical properties, and does not replace statistical evaluation. Dimensions of computational evaluation: e.g. inferential validity, algorithmic behavior, data-level plausibility, user utility, ...

Why should we care about computational evaluation of imputation methodology?

What do we care about when evaluating imputation workflows? Statistical properties such as bias/coverage/variance, empirical relevance/plausibility of the imputations, and software-engineering nitpicks like speed/convergence/fault rate. Statistical evaluation is not possible in practice, only in simulation studies. Why is statistical evaluation may not possible IRL? Bias and confidence interval coverage are impossible to determine without a comparative truth (although PPC and bootstrapping allows us to compare with respect to observed values). In this dissertation I will use computational evaluation as a suite of tools to investigate the domains of algorithmic stability, data plausibility, and user burden, next to inferential validity.

User utility does not only require statistical validity, but also usability/appropriateness/plausibility of the imputations. Even if we know the process yields correct inferences, we can evaluate the ‘fit’ of the solution.

One question to consider is: do the imputed values lie within the range of the observed values? For statistical validity this is not needed. Imputation theory does not require the imputed values to be plausible. Rubin (1976, 1987) developed the methodology to get correct population-level inferences from incomplete samples, not at the individual case-level within the sample. Bluntly stated: imputation pragmatists/purists don’t care for the imputed values themselves, as long as they result in statistically valid inferences. But end-users (e.g. clinicians) do care for plausibility. For example, imputing negative values for age will yield ‘unrealistic’ imputed values. Sometimes this is bad, such as using PMM for a variable with a sparse region, because PMM with part of sample space with too little donors might lead to bias. Sometimes this is okay, such as with polynomials (ignoring the U in imps *could* still get statistical validity).

What are other limitations of purely inferential metrics (bias, coverage)? Statistical evaluation relies on known analysis models, while FCS is substantive model agnostic. So, supplement it with computational evaluation and visual inspection. These are also possible without analysis model, which makes it more pressing today, because imputations might be used with several analyses. Not knowing the analysis model up front risks non-congeniality.

Another problem of unknown applications with potential non-congeniality is non-convergence. Without graphical or computational diagnostic for non-convergence, users might analyze imputed values of non-converged chain. For example, FCS iterations not stable, but estimand okay (because in simulation study we have true value), but what about convergence at cell level? Statistical models like MI are optimized to distinguish between effect and noise of population averages—not the individuals that make up your sample.

But also vice versa: when might the plausibility of imputed values mislead a data analyst into believing the imputations are statistically valid? When ‘looking reasonable’ is confused with ‘being consistent with the data-generating process’, such as single imputation (stochastic regression), or imputing under MCAR assumption when there is a MAR mechanism, or destroying the joint distribution of the data while preserving the marginals. Another example: in a psychological research study with baseline, some of which are missing; baseline, intervention, follow-up -> mice would inflate baseline and follow-up if both are incomplete; statistical evaluation will not show this (unbiased, coverage is okay, etc.); over-imputation or posterior predictive checks would show differences in distribution for observed and imputed data; circularity should be removed: follow-up should not predict baseline. So computational evaluation should never replace statistical evaluation.

1.4 Dissertation outline

What is the scope of this dissertation?

In this dissertation, I will focus on missing data problems with item non-response and MCAR/MAR missingness mechanisms. You might expect me to go into...

- In the category non-response: just item non-response (not unit non-response), and M(C)AR mechanisms (not MNAR). Unit non-response and MNAR missingness mechanisms are outside of the scope of this dissertation.
- In the category imputation algorithms: just FCS in the flavor of MICE (not JOMO or SMC-FCS). The missing data method under consideration is primarily multiple imputation by chained equations (MICE), but some results may generalize to other iterative (FCS) imputation algorithms. We do not cover joint modeling (JOMO) imputation or non-imputation missing data methods. Although MICE is not SMC-FCS, we will restrict the analysis of scientific interest mostly to GLM family of models.
- In the category substantive models: just GLM for statistical inference. The applications are aimed at statistical inference (not prediction or causal inference).

RQ: what tools are required to assess the quality of data-generating algorithms? in the development, evaluation, and application.

What is the status quo? Verification via statistical evaluation (but not standardized), and some validation via computational and graphical evaluation (but not systematic). Is the status quo sufficient? No, statistical evaluation is required, but not sufficient. How Rubin developed the methodology works in theory, but is not enough for contemporary practitioners.

Can we use existing tools or do we need more? and how? No, not all tools are present in the imputer’s toolbox. I contribute to the new status quo (“extending” = generalizable scientific contribution of this dissertation) by unifying how imputations are evaluated across the life cycle. There two phases: statistical properties of the method in simulation (e.g., bias and confidence-validity; statistical evaluation) & practical application (graphical and computational evaluation).

Gold standard: analytical properties. Deductive truths. But not feasible with FCS. So, implementation needed, which does require V&V.

Methodologists need standardized computational evaluations to justify new procedures, while applied researchers need accessible tools to assess whether a chosen method behaves adequately on their particular data.

- Verification of statistical properties via formal analysis (e.g. proofs) and/or simulation
- Validation of operational utility via graphical and computational evaluation

Verification

Simulation studies are verification. Assessing whether the system yields correct (known) output when a specific (known) input is supplied. Statistical evaluation is the default to evaluate the inferential validity. Examples are unit tests for software and performance assessment of methods via simulation. Verification via simulation is done, but not standardized. Verification via testing is mostly done as unit tests, but not as integration tests.

Validation

In simulation studies we can construct the “truth”, but in practice we have to rely on assumptions. Those assumptions are crucial precisely because they cannot be (fully) proven. In practice, you cannot validate whether the imputation model was appropriate, because what you would need to do so is exactly the thing you don’t have: the missing part of the incomplete data, while you only have the observed part. The thing we need is not available by definition. In practice, the best thing we can do is to rely on a combination of assumptions/theoretical justifications/substantive expertise and indirect evidence of imputation model fit: check the observed versus imputed data distributions, and inspect the imputations for signs of non-convergence. Some assumptions are impossible to definitively determine (e.g. MNAR), some we can evaluate by looking at the data. Inferential validity cannot be established.

Computational (quantitative) evaluation – metrics, tests, simulations. Graphical (qualitative) evaluation – diagnostic visualization/visual inspection. This dissertation offers insight and tools to do so. Validation tools to assess operational utility.

Chapters:

1. Evaluation paper: evaluating (new) imputation methods via simulation studies/how we know FCS works
2. Convergence paper: evaluating imputation algorithm quantitatively
3. Software: choosing/evaluating imputation models in practice

Introducing chapter 2: evaluation

This chapter argues that imputation methods should be subjected to a standardized evaluation framework for simulation studies. How to design simulation studies that are comparable? Which simulation conditions to consider, which performance metrics to report? And how to document this in such a way that others can reproduce and reuse the evaluation?

The setting where we do have the comparative truth (the true but missing values) is simulation studies. Imputers have to rely on simulation study results in order to choose the best imputation method for their incomplete variables. That's why simulation studies for imputation methodology should be comparable with one another. Then, imputers can make a 'grounded' assessment which imputation methods may be applicable, and choose the most appropriate one. Example: NRI search for lower bound, so impute as 'positive behavior', but per definition wrong statistical inf, because we intentionally bias in one direction

This chapter asks how we should judge, in a systematic way, whether imputation strategies work. **RQ: what to look out for when designing simulation studies for the evaluation of imputation methodology?**/How can computational evaluation criteria be systematically incorporated into simulation studies for imputation methodology?

- How can we trust simulation studies for imputation methods?
- How can we make them compatible & fair?

Focus on methodologists (who design imputation methods and study their properties) as opposed to imputers (applied researchers who apply imputation models to specific datasets). Methodologists need comparable, reproducible evaluations of imputation methods across scenarios. Without a shared evaluation framework, claims about 'valid, unbiased estimates' for MI might not translate to practice. How can we standardize the verification of imputation methodology?

The chapter offers advice for designing simulation studies (i.e., computational evaluation) based on literature review Liu et al. (2023).

Introducing chapter 3: convergence

This chapter investigates non-convergence in the MICE algorithm. FCS imputation, as implemented in `mice`, is an iterative procedure. Incomplete variables are imputed one-by-one, conditional on other variables. If these other variables are incomplete too, iteration is required. After the first cycle through the columns, we can get better imputations based on the now-updated imputation model predictors. In `mice`, this cycle is several times until convergence is reached (by default five iterations). Because of its iterative nature, convergence *should* be a requirement for valid inference, or is it?

This chapter investigates the questions: **How does non-convergence in iterative imputation affect inferential validity? And how can non-convergence be diagnosed?** The question is answered via a simulation study.

Introducing chapter 4: `ggmice`

This chapter introduces the software `ggmice`: an R package for building and evaluating imputation models. The `ggmice` package offers tools for the visualization of incomplete data, imputation models, and imputed data. `ggmice` supports `mice` workflows by making imputation modeling choices visible and, therefore, open to scrutiny. Compatibility with `ggplot2`'s 'grammar of graphics' makes `ggmice` graphs extendable with the familiar tools of the `tidyverse`, and adaptable for publication-ready visualizations.

Indicators for adoption: how is the software already used?

Although `ggmice` has not undergone the traditional journal-based peer review, it has been vetted in practice. The package has been released on CRAN in 2022 (Oberman et al., 2025). CRAN downloads are reaching approximately 1.000x per month at the time of writing. The accompanying Zenodo publication (Oberman, 2022) has been cited in seven scientific works. Open source code and issue tracking are available in a repository on the programming platform GitHub (github.com/amices/ggmice). There have been multiple external 'Issues' and 'Pull Requests' on GitHub, indicating that other researchers and developers have actively reviewed and improved the codebase. In short, there is active use of and involvement in `ggmice`, as indicated by regular downloads from CRAN, citations to the Zenodo record in the scientific literature, and community feedback via platforms such as GitHub and StackOverflow.

Design principles

`ggmice` is developed according to 'open source' and FAIR principles. The package is implemented in the open source language R, distributed via CRAN and GitHub, and archived on Zenodo. Its evolution is driven by user feedback (e.g., GitHub issues, StackOverflow questions, conference

discussions, teaching) and collaboration (e.g., code contributions, statistical consulting with external partners).

Other design principles are grounded in `ggmice`'s compatibility with the popular R packages `ggplot2` and `mice`. `ggplot2` offers flexible data visualizations based on the 'grammar of graphics' framework. `mice` offers flexible imputation workflows for inference with incomplete data. Together, these packages inform the design and functionalities of `ggmice`.

`ggmice` produces `ggplot` objects, leveraging the 'grammar of graphics' to create layered, easily adjustable plots. The figure that appears is the output of a series of design decisions. The grammar of graphics treats plots as structured ways of reasoning from data (Wilkinson, 2012). Layering decomposes graphics into a small set of elements. For example, variables are translated into visual properties (axes, color, etc.) via 'aesthetic mapping', and geometric objects determine the shape of the graph itself. A graphic then is a combination of several elements, and layering allows for building complex figures. The user can add or change layers without breaking the entire figure, making the graphics transparent and manipulable.

`mice` imputation models are substantive model agnostic: variable-by-variable definition of imputation models. `ggmice` supports this. Because `mice` is substantive-model agnostic, `ggmice` offers tools to inspect whether the imputation model fits the observed data, instead of imposing particular analysis models. Assumptions: imputation model should fit the observed part, imputation model should be congenial `ggmice` provides tools to see observed and imputation side-by-side, adhering to advice to evaluate the distribution of observed and imputed data (Nguyen et al., 2017)

What does `ggmice` aim to solve?

`ggmice` offers a unified toolbox for the development and evaluation of imputation models. The `ggmice` package aims to solve the following problems:

- No direct graphical comparisons between incomplete and imputed data. Visualizations with `ggmice` can be made to inspect marginal and joint distributions of incomplete variables side-to-side to compare incomplete and imputed data. Without `ggmice`, it is cumbersome to see the data before and after imputation in comparable plots. `ggmice` offers functions that visualize incomplete data distributions and their completed counterparts.
- No previously existing methods for visualizing `mice` imputation models. The structure of `mice` imputation models (i.e., predictor matrix and methods vector) are often printed to the R console as text. Or, [experientially?], exported to spreadsheets. Complex imputation models are easier to review through visualization. `ggmice` provides imputation model plots that make complex imputation specifications easier to review and discuss.
- Limited chart types and design options for imputed data. Existing plotting tools for `mids` objects not easily editable or modifiable with different chart types. Visualizations with

`ggmice` can be extended and adapted with `tidyverse` functions to create publication-ready graphs of imputed data (reflecting the uncertainty due to missingness).

Existing solutions

The `ggmice` package fills a hiatus in the software landscape, with other packages generally supporting either `mice` or `ggplot2`, but not both. Table REF provides an overview of R packages that offer visualization tools for incomplete and imputed data. In short: with `mice` you cannot plot incomplete data distributions, all other packages lack support for `mice`-imputed data, and visualization of `mice` imputation models did not exist at all prior to `ggmice`.

In particular, visualization of `mice` imputation models did not exist prior to `ggmice`. And side-by-side visualizations of incomplete and imputed data was not supported in a unified graphical framework (nl., `naniar`’s `ggplot` graphs with missing data, versus `mice`’s `lattice` plots with imputed data).

Table 1: Differential functionality of `ggmice` versus other R packages

	ggmice	mice	naniar	VIM
Incomplete data (e.g., <code>data.frame</code> object)				
– missing data patterns	+	+	±	+
– joint & marginal distributions	+	–	+	+
Imputation model building				
– associations in observed data	+	–	–	–
– associations with missingness indicators	+	+	–	–
– conditional distribution with missingness indicator	+	–	–	–
– imputation models	+	–	–	–
Imputed data (e.g., <code>mids</code> object)				
– trace plot of the imputation algorithm	+	+	–	–
– joint & marginal distributions	+	+	–	–
User utility				
– extendability (i.e., non-deterministic graph types)	+	–	±	–
– editability (i.e., design adaptable for publication)	+	–	+	–
– popularity (i.e., monthly downloads)	–	+	±	±

Note. This is an incomplete overview of the R package’s functionalities. The purpose of this table is to display *differences* in functionality based on the features offered by `ggmice`. I intend to show `ggmice`’s added value compared to existing methodology, not a universal comparison. Symbols denote high/well supported (+), medium/moderate support (±), and low/not supported (–).

Limitations

`ggmice` does not pretend that visualization yields objective truth. A good graphical fit between observed and imputed data (e.g. overlapping scatterplots) is not a guarantee of statistical validity or correct missingness assumptions. Visual diagnostics can only suggest plausibility or highlight obvious misfits (imputation models should fit the observed part of the data). They cannot determinately identify the missing-data mechanism. The interpretation remains subjective (e.g. convergence, MAR assumption, etc.). Visualization is not a replacement for, but support to evaluation. Also check metrics such as FMI.

Future research & development

Planned extensions further align with the philosophy of making imputation workflows and their consequences visible. Some directions for future development include:

- Visualizing `mira` objects (analysis results across multiple imputations).
- Adding posterior predictive checks and over-imputation plots.
- Quantifying uncertainty at the pattern or cell level.
- Tools for sensitivity analyses (e.g. inducing MNAR structure in the observed data and inspecting effects on imputations and inferences).
- Investigating which data-level diagnostics are actually informative about imputation model adequacy in practice (i.e., when inferential validity cannot be evaluated).

In short, `ggmice` is an attempt to make imputation model assumptions and consequences as transparent as possible.

2 Chapter 2: evaluation

[TODO: link/insert + credit]

REVIEW ARTICLE

Toward a standardized evaluation of imputation methodology

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This article has earned an open data badge “**Reproducible Research**” for making publicly available the code necessary to reproduce the reported results. The results reported in this article could fully be reproduced.

Abstract

Developing new imputation methodology has become a very active field. Unfortunately, there is no consensus on how to perform simulation studies to evaluate the properties of imputation methods. In part, this may be due to different aims between fields and studies. For example, when evaluating imputation techniques aimed at prediction, different aims may be formulated than when statistical inference is of interest. The lack of consensus may also stem from different personal preferences or scientific backgrounds. All in all, the lack of common ground in evaluating imputation methodology may lead to suboptimal use in practice. In this paper, we propose a move toward a standardized evaluation of imputation methodology. To demonstrate the need for standardization, we highlight a set of possible pitfalls that bring forth a chain of potential problems in the objective assessment of the performance of imputation routines. Additionally, we suggest a course of action for simulating and evaluating missing data problems. Our suggested course of action is by no means meant to serve as a complete cookbook, but rather meant to incite critical thinking and a move to objective and fair evaluations of imputation methodology. We invite the readers of this paper to contribute to the suggested course of action.

KEYWORDS

evaluation, imputation, missing data, simulation studies

1 | INTRODUCTION

Imputation is a state-of-the-art technique for drawing valid conclusions from incomplete data. The technique has earned a permanent spot in research and policy-making, demonstrated, for example, by the detailed manual created by the National Research Council (Little et al., 2012). Although top-down enforcement of valid ways to handle missing data is not yet very pronounced, an increasing amount of researchers and data scientists are embracing and developing imputation methodology. After all, the principle of imputation is very intuitive. The idea behind imputation is to impute (fill in) missing values to obtain a valid estimate of what could have been. The completed data that are thus obtained can be analyzed by standard techniques, which conveniently separates the missing data problem from the analysis.

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For some data analyses, a single imputation may suffice, while for other data analyses, multiple imputations are needed. For example, in the case of inferential analyses, such multiple imputations (Rubin, 1976) have proven to be a valuable technique for obtaining valid inferences on incomplete data sets. The resulting multiple inferences on multiple completed data sets can be combined into a single inference using Rubin's rules (Rubin, 1987, p. 76). When generating predicted values is the goal of the data analysis, a single imputation may already generate sufficient predicted values (Nijman et al., 2020).

In all scenarios, it can be said that the quality of the solution obtained by imputation depends on the statistical properties of the incomplete data and the degree to which the imputation procedure is able to capture these properties when modeling missing values. In general, it holds that modeling missing data become more challenging when the amount of missingness increases. However, when (strong) relations in the data are present, the observed parts can hold great predictive power for the models that estimate the missingness. In that case, imputation may be substantially more efficient than the ubiquitous complete case analysis.

When evaluating the statistical properties—and thereby the practical applicability—of imputation methodology, researchers most often make use of simulation studies. Such studies are typically composed of an analysis pipeline in which data are subsequently generated, made incomplete, and imputed repeatedly under different simulation conditions. A set of evaluation criteria is then postulated to evaluate the performance of one or more missing data methods. Recent attention on best practices for method evaluation by means of simulation (Morris et al., 2019) and Pawel et al. (2022)'s advice to preregister simulation protocols will guide methodologists in conceptualizing their simulation workflow. However, there is no consensus on how to design, execute, and report simulation studies aimed to evaluate imputation methodology. Without a “gold standard” for evaluation, the validity and comparability of simulation setups may differ tremendously from one developer to another. This brings forth a chain of potential problems in the objective assessment of imputation method performance within and across studies, which could lead to suboptimal use of imputation in practice.

The purpose of this paper is threefold: First, we raise some concerns with respect to evaluating imputation methodology. These concerns stem from careful consideration with fellow “imputers” and from encounters as reviewers for statistical journals. Second, we provide imputation methodologists with a suggested course of action when using simulation studies to evaluate imputation techniques for missing data problems. This suggested approach should identify common ground, but is in no way intended as an absolute solution. This identifies the third purpose of our paper: discussion. We hope to elicit critical thinking regarding the problems at hand. We are all convinced that our methodology has some merit. But for sake of progress, it would be much more advantageous if the aim of our evaluations would go beyond *proving the point* and would legitimately consider the statistical properties.

2 | WHY SOME EVALUATIONS SHOULD NOT BE TRUSTED

The ideal evaluation of imputation methodology may differ between studies. A simulation study developed for the comparison of several imputation methods will typically have a different design than one aimed at establishing the inferential validity of a single (novel) method. How the evaluated methods are used in practice thereby governs the simulation procedure. Since the simulation aims and setup are intertwined with the choice of the evaluated imputation methods, one should always assess the suitability of the chosen evaluation metrics with the imputation methods' purpose in mind.

We limit the scope of this paper to comparative simulation studies in the context of statistical inference. With that, we exclude the evaluation of imputation methodology for detailed aspects of causal inference and prediction. The type of simulation study under consideration in this paper requires a form of “comparative truth” to assess the inferential validity of imputation methods. This is in contrast to simulation studies aimed at comparing the predictive performance of imputation and prediction method pairs. Such a simulation design typically does not start out from a “ground truth” (e.g., a complete data set in which missingness is subsequently induced by the simulator). Rather, the imputation and prediction method pairs are evaluated on their ability to yield high predictive accuracy in one or more incomplete benchmark data sets (Liu et al., 2021). In these designs, only the comparative performance of methods may be established. We do not recommend this approach if the inferential validity of the imputations is of interest. For an overview of missingness in the prediction context, see, for example, Wood et al. (2015) and Sperrin et al. (2020). Missing data within a causal inference framework have been described by, for example, Moreno-Betancur et al. (2018) and Mohan and Pearl (2021).

Within the scope of comparative simulation studies aimed at statistical inference, we observe several problems. To demonstrate the broad impact of these problems, we recognize the following four distinct categories: problems with simulation design, problems with data generation, problems with missingness generation, and problems with performance evaluation. We further detail the impact each of these problems may have on the validity of evaluations.

2.1 | Simulation design

Before setting up a simulation study, the simulator should clearly define and describe the scope of their evaluations. If the simulation parameters are unclear, any methodological problems with simulation study design may be obfuscated by the cognitive problem of generalization (Greenland, 2017). For example, misinterpretation and (unintended) questionable research practices can prompt extrapolation beyond the scope of the simulations (Greenland, 2017; Pawel et al., 2022). Imputation methods that seem to work well in simulations may then not be suitable in practice, when applied to incomplete empirical data.

One potential reason for discrepancies between imputation method performance in simulations versus applications is the use of sampling variance in the simulation design. For example, when dealing with infinite populations, the conventional pooling rules proposed by Rubin (1987, pp. 76–77) apply. When dealing with a finite incomplete population, some adjustment is needed for pooling rules to yield correct inferences. In such cases, valid inferences can be obtained by excluding the sampling variance components from the calculation of the total variance about $(Q - \bar{Q})$ (Raghunathan et al., 2003; Vink & Van Buuren, 2014), where Q would be the estimand and $\bar{Q} = \sum_{l=1}^m \hat{Q}_l / m$ is the average over the m imputed estimates (cf. Rubin, 1987, p. 76–77). Vink and Van Buuren (2014) propose that the total variance $T = B + B/m$ would then solely rely on the between imputation variance $B = \sum_{l=1}^m (\hat{Q}_l - \bar{Q})'(\hat{Q}_l - \bar{Q}) / m - 1$ and demonstrate that the degrees of freedom would then equal $m - 1$.

We can make use of this unique real-world data property when simulating the performance of incomplete data methodology. When we take a single drawn complete data set as our comparative truth, just inducing simulated missingness would suffice, and the necessary Monte Carlo variance for our evaluations would stem from the sampled missingness. The above-detailed pooling rules can be used to obtain valid inferences in those simulations. A detailed overview of simulation strategies that adopt this approach, as well as conventional simulation strategies can be found in Vink (2022).

We would like to note that it is not necessary to exclude sampling variance from missing data simulations, as the additional variance induced by the sampling step would be ignorable. In practice, however, we have found it to be convenient to allow for a single data set to serve as comparative truth in all simulation repetitions. Especially for data-driven simulation strategies with complex data transformations, it can be challenging to derive the true parametric references to evaluate against. Evaluating against a single generated set would then avoid many procedural problems and computational challenges. In practice, there are also other benefits to this simulation approach as it is computationally convenient, requires less data archiving, and may potentially sharpen comparisons between imputation methods, thereby clarifying inconclusive results on method imputation performance. After all, we are interested in solving for the missingness and can do without the ignorable noise induced by the sampling mechanism for evaluation in such studies.

A well-designed simulation does not guarantee fair evaluations. Estimands and other simulation targets should be clearly and unambiguously defined in the context of the study aim (Petersen & Van der Laan, 2014). The study aim and the required level of precision may also inform the number of simulation repetitions (e.g., as determined from a maximum tolerable level of uncertainty in terms of a performance measure's Monte Carlo error; see Morris et al., 2019). The simulation design should ideally be fully factorial, that is, varying each simulation condition against all other conditions. There may, for example, be interactions between the source and amount of missingness in the incomplete data and the efficacy of imputation methods—that is, the validity of the assumed missingness mechanism becomes increasingly important with higher missingness proportions (Schouten & Vink, 2021). If there is more than one imputation method under evaluation, the simulation design should apply each method to every incomplete data set. Applying all methods to the same incomplete data set is computationally convenient, and minimizes unnecessary variation, which makes for fairer comparisons.

2.2 | Data generation

To evaluate the ability of an imputation routine to handle missingness, a form of ground truth has to be established. Those who perform simulation studies are in the luxury position to establish the truth beforehand by choosing a data-generating mechanism. Data-generating mechanisms define how a complete dataset is obtained at the start of each simulation repetition. We highlight two general data generation approaches: (i) model-based simulation, in which data are drawn from a known statistical model or probability distribution, such as the multivariate normal distribution; and (ii) design-based simulation, where data are sampled from a sufficiently large observed set, such as official registers. Model-based data-

generating mechanisms have advantages in flexibility and precision, because data are generated from a known statistical model and the true theoretical parameters can be derived. A design-based approach is often used in situations where a probability distribution is not available, or where real-life data structures are of interest. Of course, the simulator should choose their data-generating mechanism in line with the study scope. Other methods to simulate data may be more suitable, for example, synthetization of the actual data-generating process (Li et al., 2022; Volker & Vink, 2021).

The challenge with model-based data-generating mechanisms is that a method's performance on simulated data may not translate to empirical data. Real-life data may not follow a known theoretical distribution, so there is no guarantee that simulation results are generalizable. Moreover, data are often generated such that the problem being studied is most pronounced, for example, with consistently high correlations between groups of variables. This results in simulated data that contain such valuable information structures that no matter what type of missingness would subsequently be induced, the observed parts of the data will still hold much (if not all) of the information about the missing part. Unsurprisingly, the performance of any imputation method will then be evaluated as good.

Another threat to the generalizability of model-based simulations is the use of a single model for both data generation and imputation. If data are generated following a model that is also used for imputing the data, the imputation approach will be deemed good (or better than other methods) purely due to the evaluated conditions being in favor of the problem that is studied. Other potentially unfair comparative advantages in favor of a certain imputation method may occur due to characteristics of the generated data, such as the number of observations, the number of variables, the measurement level, and the coherence between variables. In contrast to design-based studies, such characteristics are not always explicit simulation conditions, which may give a false sense of objectivity.

An obvious problem with design-based simulation is that obtaining a large data set without missing entries can be challenging. Most real-world data contains at least some missing entries, for which the true underlying missing data model is—by definition—unknown. Therefore, the simulator needs to deal with missingness in *some* way before incomplete empirical data can serve as comparative truth in the simulations. It might seem like an intuitive solution to only draw complete cases from the large data set, which would indeed yield complete samples. However, there may be inherent differences between cases with and cases without any missing values, due to the unknown missing data model. Only sampling complete cases from the data set may thus result in samples that fail to capture all relevant real-world conditions, which, in turn, refutes the main reason for using design-based simulation. Another way to deal with missingness in a design-based simulation is to impute the incomplete data set once, to obtain a single completed data set to draw samples from. Unfortunately, this practice may favor the imputation method that was used in this initial imputation step throughout any further evaluations. Just to be clear: leaving the missingness as-is and inducing additional missing values to impute is no option here, because we would not have a real and unbiased comparative truth.

2.3 | Missingness generation

Often, reports of simulation studies remain vague about the missingness conditions under investigation and, even worse, some authors only report something like:

We generated missing data following a missing at random missingness mechanism.

This should be considered unacceptable, as claims about the validity of the imputation inference heavily depend on the simulated missingness conditions, such as missingness mechanisms and missingness patterns. Missingness mechanisms describe the relationship between missing entries and observed data values, whereas missingness patterns concern the location of missing entries across incomplete data (Little & Rubin, 2020, p. 8). Under this definition, there is a row-wise element to the missingness pattern describing which variables are jointly observed, and a column-wise element encompassing the amount of missingness in the data.

Even the terminology on missingness generation can be confusing. Does a missingness proportion of 50% mean that half of the entries in an incomplete data set are missing, or that half of the rows have at least one missing entry? In this paper, we will henceforth refer to the latter as the proportion of incomplete cases, and keep the term “missingness proportion” restricted to the variable-by-variable interpretation. The distinction between the two concepts, however, is not always clear in the literature, and convoluting the terms may lead to incorrect generalizations, because they rarely mean the same (e.g., a proportion of 50% incomplete cases in bivariate data could translate to a missingness proportion of 25% in both variables, or one completely observed variable and one with 50% missingness). Simulators should therefore

be explicit in their description of the missing data-generating model and should make sure that the model fits the aim of their simulations.

Extrapolating simulation results from a specific missing data-generating model to more intricate (empirical) missingness could lead to suboptimal advice in practice. For example, if a simulator induces missingness in the outcome variable of their analysis model exclusively, they may inadvertently induce missingness according to the special case described by Van Buuren (2018, §2.7). Under this specific missingness model, list-wise deletion may outperform any imputation method. Such a conclusion would, unfortunately, only translate to data that adhere to the same special case, while biasing inferences in other cases. Another example of potential side effects of missingness generation may occur when multivariate missingness is generated using step-wise univariate missingness induction. The resulting incomplete data may then not have the desired statistical properties or unintentional nonignorable missingness may be generated (Robins & Gill, 1997; Schouten et al., 2018). This complicates any definitive conclusions about imputation method performance in relation to the data generation parameters.

Missingness should be induced according to several sets of missing data conditions. We encourage simulators to consider different missingness patterns and mechanisms. Missingness mechanisms were first defined in Rubin's seminal work (Rubin, 1976). For more recent considerations, see, for example, Doretti et al. (2018), Little and Rubin (2020), Moreno-Betancur et al. (2018), Mohan and Pearl (2021), Mealli and Rubin (2015), Schomaker (2021), Seaman et al. (2013), and Schouten and Vink (2021). Although not every missingness mechanism is realistically assumed in practice, they can all offer valuable insights as simulation condition. A disconnect between induced missingness, real missingness, and assumed missingness may result in simulation studies that are not as informative as they could be.

Truly random missingness across all data entries ("missing completely at random" or MCAR under Rubin's (1976) definition) may be considered as a necessary simulation condition for the evaluation of imputation procedures, because the statistical properties of the observed data given the missing data are known. Any imputation routine that cannot at least mimic the performance of the observed data inference should be deemed inefficient in the scope of the simulation. If an imputation method is not able to solve the problem (i.e., yield valid inference) under MCAR, the statistical properties of the procedure are not universally sound. Sadly, the straightforward case of MCAR is often neglected in simulation studies, whereas it offers an informative reference condition in many simulation setups.

Besides MCAR missingness, observed-data-dependent missingness mechanisms—such as missing at random (MAR) missingness—are paramount in statistical inference investigations. A straightforward technique for inducing univariate MAR missingness is described in Van Buuren (2018, §3.2.4), whereas generalizations to multivariate MAR missingness can be found in Schouten et al. (2018). If the missingness is to be induced in longitudinal data, generating the missingness through autoregressive MAR models can be useful (see, e.g., Shara et al., 2015, models 2 and 3). It is advisable to investigate varying shapes of MAR missingness to achieve a more realistic indication of the robustness of the imputation performance across the range of random missingness. The effects of different types of MAR mechanisms are described in Schouten and Vink (2021).

Inducing an MAR mechanism presents the simulator with a choice, namely, the type or functional form of the missingness model. Given the simulated data distributions, one random missingness model may be far more disastrous to the observed information than another model (Schouten & Vink, 2021). This may influence the performance of some (but not necessarily all) imputation routines. For example, inference from hot-deck techniques such as predictive mean matching (Little, 1988; Rubin, 1986) may be more severely impacted by large amounts of one-tailed missingness than inference from parametric techniques. It would be a shame to overlook such results due to the focus on a single functional form of the missingness-generating model.

Although MAR missingness is often considered as a simulation condition, the problem of spurious MAR is generally overlooked. With MAR missingness mechanisms, observed relations in the data are used to induce missingness during simulation (e.g., weight is made incomplete based on observed gender to mimic a situation wherein one gender is less likely to disclose their weight). These relations in the observed data may, however, be weak or nonexistent. If MAR is induced based on weaker relations in the true data, claims for a method's applicability to situations where the missingness is random become less valid. The most extreme example would be when MAR is induced from data without multivariate relations. The inferential implications of the missingness would then mimic those of MCAR, even though the missingness is generated randomly. Figure 1 demonstrates this: Both univariate MAR mechanisms in Figure 1 are induced based on the right tail, but only the mechanisms for which the *variable to ampute* and the *missingness covariate* have a positive correlation will yield valid MAR mechanisms. In the case of zero correlation between the data columns, iterative univariate non-MCAR missingness schemes may induce invalid missingness mechanisms that mimic MCAR. Table 1 demonstrates this and highlights the importance of including complete case analysis as an evaluation analysis to allow for identifying

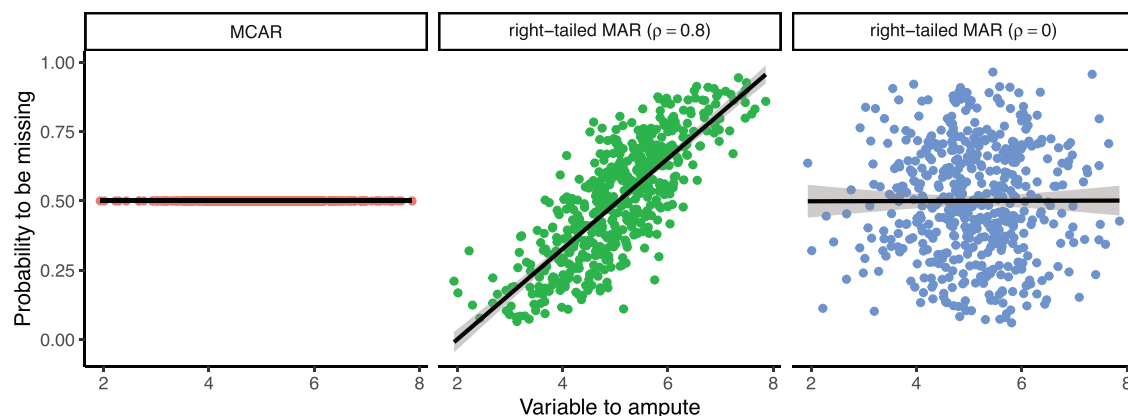


FIGURE 1 Three different realizations of missingness generation for MCAR and right-tailed MAR missingness mechanisms for different levels of correlation ρ between the *variable to ampute* and the *missingness covariate* that guide the probability to be missing. Note that the conditional probability of right-tailed MAR ($\rho = 0$) mimics that of MCAR because the variable to ampute and the missingness covariate have $\rho = 0$.

TABLE 1 Inferences obtained by complete case analysis (CCA) and multiple imputation over 1000 simulations. Displayed are the bias of the mean, coverage rate (cov) of the corresponding 95% confidence interval, and the confidence interval width (ciw). Imputations are generated by Bayesian linear regression imputation.

Mechanism		CCA			Imputation		
		Bias	Cov	Ciw	Bias	Cov	Ciw
$[\rho = 0.8]$	mcar	0.000	0.940	0.178	−0.000	0.941	0.160
$[\rho = 0.8]$	mar	0.335	0.000	0.169	0.004	0.954	0.190
$[\rho = 0.0]$	mar	0.001	0.973	0.180	0.011	0.939	0.311

such invalidly generated missingness mechanisms in simulation. In fact, this would amount to one of the special cases described by Van Buuren (2018) under which complete case analysis would be more efficient than imputation: the missingness does not depend on the incomplete variable. This property might be useful in practice, but considering it as a condition to evaluate the performance under MAR missingness is pointless.

A nonignorable or MNAR mechanism—where the unobserved values themselves may also play a role in the probability to be missing—might fall outside many studies' scope, yet could yield informative insights for daily practice. Even if an imputation method is specifically developed with ignorable missingness in mind, chances are that the method will be applied to nonignorable empirical missingness at some point in time. After all, for every MNAR mechanism, there is an MAR mechanism with equal fit and one cannot definitively verify that empirical missingness is random (Molenberghs et al., 2008). Therefore, it can be argued that MNAR is the more likely mechanism for real-life missingness scenarios. It may be wise to include MNAR missingness in the simulation study just in case: A method that performs well under MAR and some cases of MNAR may be preferable over a method that only works under MAR, and not under MNAR. Such “late-phase” methodological investigations (see Heinze et al., 2022) can provide an understanding of an imputation method's robustness. Alternatively, a sensitivity analysis may provide an indication of the validity of the obtained inference, given that the assumed missingness mechanism is suspected to be invalid (see, e.g., Molenberghs et al., 2014, part 5).

In addition to varying the missingness mechanisms, each simulated mechanism should be combined with different missingness patterns. Remember that missingness is only ignorable under MAR when the parameter of the data is distinct and *a-priori* independent from the parameter of the missing data process (Little & Rubin, 2020, Corollary 6.1A). Under MAR missingness, we assume that we may use the observed data to make inferences about the joint (observed and unobserved) data. The dependency of the procedure on the assumption under which we obtain inference is only influenced by the amount of missingness. If there is no missingness—or if there is no data, for that matter—the inference does not depend on the assumption. Alternatively, the validity of assumptions becomes increasingly important when the missingness increases. Since we control the MAR mechanism, the assumption under which we may solve the missing data problem should hold and it is only fair to assess performance under stringent missingness conditions. We therefore propose to evaluate imputation methodology under several missingness proportions to emulate a realistic range in sever-

ity of the missingness problem. Depending on how the missingness mechanism interacts with the simulated data, higher missingness proportions may yield biased or invalid completed data inferences. The missingness proportions should thus be considered carefully.

The above emphasizes the need for both thorough evaluation and thorough documentation of the intended and used missingness mechanisms. Simulators must make clear which mechanisms are of interest. To be able to verify these mechanisms, simulators must also be transparent about the process that generated the missingness. For example, by sharing the code or detailing the process that generated the missing values. Alternatively, missingness generation devices, such as the `mice::ampute()` function, can be used to generate valid, pattern-based missingness in any structured data set (Schouten et al., 2018; Van Buuren & Groothuis-Oudshoorn, 2011). Finally, simulators must include simulation conditions and evaluation parameters that make it possible to identify the proper generation of missing values. This would prevent that errors in the missingness generation would go unnoticed.

2.4 | Performance evaluation

The evaluation criteria used to assess imputation performance vary from one simulator to another. This is not surprising as people from different fields could have a different focus on the problem at hand. There are, however, some overarching issues when assessing imputation method performance. In the first place, there are pitfalls in the evaluation of the estimates that are obtained by fitting the analysis model after imputation. For example, focusing on one performance measure over another may yield different conclusions about imputation efficacy. In the second place, diagnostic evaluation of the generated imputations is often left out of simulation study results. Identifying problems with the imputation-generating process (e.g., an iterative imputation algorithm) may offer explanations for underperformance of imputation methods. Such evaluations, however, are typically omitted and valuable insights into the imputation method(s) may then be overlooked.

At minimum, imputation method performance should be quantified using appropriate measures. It depends on the specifics of each simulation study which performance measures would be most suitable. If the goal is inference—not prediction—the uncertainty about estimates needs to be properly quantified. To capture this uncertainty, the standard errors of the estimates should be correctly calculated, which requires multiple imputation. The aim of multiple imputation is not to reproduce the data, but to allow for obtaining valid inference given that the data are incomplete. This means that, given the framework provided by Rubin (1987), and for any $(Q - \bar{Q})$, statistical properties such as bias, confidence intervals, and the coverage rate of the confidence intervals should always be studied. We therefore recommend evaluating the following points:

- (i) The methods should preferably be unbiased. Note that the way bias is calculated should be carefully chosen and described, because this can greatly influence the interpretation of the results. A negligible absolute bias, for example, may already yield an almost infinite relative bias if the true value of the estimand is zero. In most cases, unbiased estimation may be expected under an MCAR missingness mechanism, which can be easily verified by including MCAR missingness in the scope of missingness investigations.
- (ii) The intervals around estimates should have a valid coverage of the population (i.e., true) value. Coverage of a 95% interval should in theory be $\geq 95\%$, where a coverage rate of 95% would be most efficient (Neyman, 1934, p. 589). Undercoverage indicates that the procedure is not confidence valid and may lead to invalid inference. Undercoverage may occur when the estimation procedure is either biased, too liberal leading to intervals that are too narrow, or both. Overcoverage, though technically not confidence invalid, would indicate that efficiency could still be gained. In such cases, a narrower interval exists for which the method would still yield confidence valid result.
- (iii) The width of the confidence or credible interval may convey statistical efficiency, which should be considered to compare imputation methods. Wider intervals are associated with more uncertainty, whereas a more narrow interval that is still properly covered indicates a sharper inference. However, inference from a wider interval that is properly covered is to be considered more valid than a more narrow interval that is not properly covered anymore. That said, with valid nominal coverage, the method that yields the narrowest interval would be most efficient.
- (iv) Resemblance to the true data values may be quantified using the root mean squared error (RMSE). We do not generally recommend the use of the RMSE as evaluation criterion, because this metric does not account for the inherent uncertainty of the missing values and it may inflate the type I error rate of statistical inferences (Chapter 2.6 Van Buuren, 2018). However, if a study is aimed at obtaining predictions, and inferential validity is not of interest, the

RMSE of the prediction errors (residuals) can be compared between methods. Then, the RMSE may yield valuable information about the methods' comparative efficiency in terms of accuracy and precision.

After establishing the statistical validity of an imputation routine on the level of the estimand, it remains paramount to also study the imputation-generating process. Omitting such investigations may yield suboptimal results. One simple example arises often in practice: generating implausible imputed values. Even though the estimation on the analysis level may be justified, some methods can yield imputations that may seem completely invalid to applied researchers. For example, one could very accurately estimate average human height by filling in negative values and values that are unrealistically large. While the obtained inference could still be valid under such imputations, the plausibility of the imputed values given the observed data should be under scrutiny. Under many circumstances, imputation methods may be realistically expected to preserve both marginal and conditional distributions with respect to the comparative truth. Imputation methods that fail to do so should not be considered general-purpose methods. Many techniques can yield valid inferences, but techniques that sample realistic or plausible values may be preferable in practice.

The evaluation of simulated results goes far beyond plausibility and distributional characteristics. The quality and validity of an imputation task will also rely on the relation between the missing data models, imputation models, and analysis models. When a joint model exists whose conditionals would include the imputation and analysis model, then the imputation model and analysis are said to be congenial (Bartlett et al., 2015; Meng, 1994). The congeniality of all imputation models should be assessed, but methods to assess the suitability of imputation models and diagnose misfit often rely on visual inspection of the imputations (see, e.g., Abayomi et al., 2008; Bondarenko & Raghunathan, 2016). This makes its assessment a challenging and time-consuming endeavor in many simulation studies.

When algorithms are used, the simulator should study the validity of the algorithmic process. Many contemporary imputation techniques rely on iterative algorithms, such as the Gibbs sampler, to generate imputations. As with any iterative algorithm—but especially with imputation algorithms that are critically considered to be possibly incompatible Gibbs samplers (PIGS, Li et al., 2012)—algorithmic convergence should be carefully evaluated. Oberman et al. (2021) demonstrate that statistical validity can happen before algorithmic convergence is reached, but this is not a guarantee. Unfortunately, there is no universal quantitative method to diagnose nonconvergence in iterative imputation algorithms (Oberman et al., 2021; Zhu & Raghunathan, 2015) and the alternative (visual inspection of the imputation algorithm; Van Buuren, 2018) is neither efficient nor fail-proof. As a result, imputation algorithms may be terminated before reaching a stable state, which could yield suboptimal imputations and underestimated performance of the method. Problems with producing stable imputations are not exclusive to iterative imputation algorithms. Any method may run into failures of the imputation-generating process and subsequently lack results, for example, due to overparameterization errors.

Every imputation workflow should therefore contain an evaluation of the obtained imputations. Even though inspecting each imputation may be labor-intensive due to the number of imputations generated in a simulation study, we highly recommend simulators consider the following aspects:

- (i) The absence of nonconvergence in the imputation-generating process is a minimum requirement for any imputation method. If nonconvergence is suspected, the inference resulting from the imputations might be invalid. However, preliminary work suggests that iterative imputation algorithms could achieve inferential validity before reaching a stable state (Oberman et al., 2021).
- (ii) The fit of the imputation model may be verified with the help of a posterior predictive check (Nguyen et al., 2017; Zhao, 2022). A straightforward posterior predictive check for imputation methodology is the multiple overimputation of observed data values (Cai et al., 2022). If the statistical properties of the overimputed values are equivalent to those of the observed data values, one could infer that the imputation model fits the observed part of the incomplete data reasonably well. Then, by extension, one could assume that the imputation model might be able to produce good imputations for the missing part of the data too. This overimputation procedure is straightforward with the `overimpute()` function in *Amelia II* (Honaker et al., 2011) and the `where` argument in *mice* (Cai et al., 2022; Van Buuren & Groothuis-Oudshoorn, 2011).
- (iii) The distributional characteristics of the imputations should be inspected for anomalies. The distribution of the incomplete data may differ greatly from the observed data. Under anything but the MCAR assumption, this can be expected. When evaluating imputations, the distributional shapes should be checked and diagnostic evaluations should be performed (see Abayomi et al., 2008, for a detailed overview of diagnostic evaluation for multivariate imputations). When anomalies are found, and if the imputation method is valid, there should be an explanation, especially in the controlled environment of a properly executed simulation study.

- (iv) Finally, the plausibility of the imputed values may be evaluated. Plausible imputations—imputations that could be real values if they had been observed—are not a necessary condition for obtaining valid inference. However, in practice, especially when the imputer and the analyst are different persons, plausible imputations may be a desired property. One would prefer an imputation technique to yield both valid inference and plausible imputations. It should be studied if an imputation method is prone to deliver such impractical results, and if so, under what conditions. When evaluating imputation routines, the evaluator should mention whether the routine is prone to deliver implausible values. We must note, however, that the aim of imputation should be valid inference in the first place, and plausibility in the second place.

There are parallels between evaluating the performance and evaluating the missingness. Just like using MCAR as a condition to benchmark the validity of the missing data solution, we can use the simulation's complete data (i.e., without missing values) as an upper limit for our performance evaluation. In addition, the simulator should also perform complete case analysis, that is, analysis on the incomplete data without solving for the missingness. We know the theoretical properties of complete case analysis, which makes the ad hoc technique useful as a lower limit for evaluating imputation performance. Complete case analysis may therefore serve as a benchmark method against which imputation performance should be evaluated. Realistically, the simulator would expect the imputation solution to perform not worse than the complete case analysis and, preferably, mimic the complete data results. Moreover, pairing complete case analysis with MCAR and MAR mechanisms allows the simulator to evaluate the impact of the missing data generation process under the chosen analysis model and to identify the unintentional generation of accidental MNAR.

Last, when the simulator is on the verge of drawing conclusions about the performance of the imputation methodology, the performance should be carefully qualified. Comparing the performance of an imputation routine, given a population (or true) parameter allows for quantitative evaluation. Yet, in order to pose qualitative statements about the performance on simulated conditions, comparative methodology is required. For example, when claiming that imputation performance is unacceptable when deviations from normality become rather stringent, such performance is highly dependent on the simulation conditions that are used. For a well-balanced judgment about the severity of the performance drop, comparative simulations with, for example, nonparametric models should be executed. A method may perform badly, but if it still outperforms every other approach, it may yet be of great practical relevance.

3 | SUGGESTED COURSE OF ACTION

We encourage simulators to carefully consider and document their choices in the evaluation of imputation methodology. Simulation studies should not only be well executed, but also properly reported. For example, the inconsistent display of simulation conditions may impact the objectivity of metaevaluations over imputation methods, as one method's performance may appear to be favorable because of less stringent simulation conditions. This may ultimately lead to statisticians recommending a less efficient method to applied researchers, thereby limiting the efficiency of the imputation approach and unnecessarily lowering the statistical power. Simulators should therefore be explicit in the descriptions of their evaluations. However, deciding what and how to report may be challenging. The simulation design could be presented textually, in a flowchart or as a block of pseudocode, whereas missingness mechanisms could be written as a function of the data or displayed graphically. Ideally, the evaluations should be supplemented by an online repository with all of the data and code required to reproduce the simulation results. To aid simulators in reporting and move toward standardization in evaluation, we provide a draft version for reporting guidelines in Appendix A.1 (also available from www.gerkovink.com/evaluation). We invite the readers of this paper to contribute to its development.

We aim to elicit critical thinking about incomplete data simulation and to establish a common ground for the evaluation of imputation routines. Such a common ground would be the basis of a standardized evaluation. This would allow for fairer and more efficient comparisons between imputation techniques. Ultimately, it would be desirable to evaluate every imputation routine against the same standardized set in order to quantify the statistical properties across imputation routines. If properly executed, such evaluations would allow for careful matching of imputation methodologies to new missing data problems.

ACKNOWLEDGMENTS

We thank the Amices team for the fruitful discussions and highly value the comments and suggestions from the reviewers and the editors.

CONFLICT OF INTEREST STATEMENT

The authors have declared no conflict of interest.

DATA AVAILABILITY STATEMENT

This review article only contains simulated data, which is openly available from <https://github.com/gerkovink/evaluation>

OPEN RESEARCH BADGES

 This article has earned an Open Data badge for making publicly available the digitally-shareable data necessary to reproduce the reported results. The data is available in the [Supporting Information](#) section.

This article has earned an open data badge “**Reproducible Research**” for making publicly available the code necessary to reproduce the reported results. The results reported in this article could fully be reproduced.

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REFERENCES

- Abayomi, K., Gelman, A., & Levy, M. (2008). Diagnostics for multivariate imputations. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 57(3), 273–291.
- Barnard, J., & Rubin, D. B. (1999). Miscellanea. small-sample degrees of freedom with multiple imputation. *Biometrika*, 86(4), 948–955.
- Bartlett, J. W., Seaman, S. R., White, I. R., Carpenter, J. R., & Initiative*, A. D. N. (2015). Multiple imputation of covariates by fully conditional specification: Accommodating the substantive model. *Statistical Methods in Medical Research*, 24(4), 462–487.
- Bondarenko, I., & Raghunathan, T. (2016). Graphical and numerical diagnostic tools to assess suitability of multiple imputations and imputation models. *Statistics in Medicine*, 35(17), 3007–3020.
- Cai, M., Van Buuren, S., & Vink, G. (2022). Graphical and numerical diagnostic tools to assess multiple imputation models by posterior predictive checking. *arXiv preprint. arXiv:2208.12929*. <https://arxiv.org/abs/2208.12929>
- Doretti, M., Geneletti, S., & Stanghellini, E. (2018). Missing data: A unified taxonomy guided by conditional independence. *International Statistical Review*, 86(2), 189–204.
- Greenland, S. (2017). Invited commentary: The need for cognitive science in methodology. *American Journal of Epidemiology*, 186(6), 639–645.
- Heinze, G., Boulesteix, A.-L., Kammer, M., Morris, T. P., & White, I. R. (2022). Phases of methodological research in biostatistics - Building the evidence base for new methods. <https://arxiv.org/abs/2209.13358>
- Honaker, J., King, G., & Blackwell, M. (2011). Amelia II: A program for missing data. *Journal of Statistical Software*, 45(7), 1–47. <https://www.jstatsoft.org/v45/i07/>
- Li, F., Yu, Y., & Rubin, D. B. (2012). *Imputing missing data by fully conditional models: Some cautionary examples and guidelines*. Duke University Department of Statistical Science Discussion Paper, 1124.
- Li, H., Rosete, S., Coyle, J., Phillips, R. V., Hejazi, N. S., Malenica, I., Arnold, B. F., Benjamin-Chung, J., Mertens, A., & Colford Jr, J. M. (2022). Evaluating the robustness of targeted maximum likelihood estimators via realistic simulations in nutrition intervention trials. *Statistics in Medicine*, 41(12), 2132–2165.
- Little, & Rubin (2020). *Statistical analysis with missing data* (3rd ed.). Wiley Series in Probability and Statistics. Wiley.
- Little, R. J. (1988). Missing-data adjustments in large surveys. *Journal of Business & Economic Statistics*, 6(3), 287–296.
- Little, R. J., D’Agostino, R., Cohen, M. L., Dickersin, K., Emerson, S. S., Farrar, J. T., Frangakis, C., Hogan, J. W., Molenberghs, G., Murphy, S. A., Neaton, J. D., Rotnitzky, A., Scharfstein, D., Shih, W. J., Siegel, J. P., & Stern, H. (2012). The prevention and treatment of missing data in clinical trials. *New England Journal of Medicine*, 367(14), 1355–1360.
- Liu, D., Oberman, H. I., Muñoz, J., Hoogland, J., & Debray, T. P. A. (2021). Quality control, data cleaning, imputation. *arXiv preprint arXiv:2110.15877*.
- Mealli, F., & Rubin, D. B. (2015). Clarifying missing at random and related definitions, and implications when coupled with exchangeability. *Biometrika*, 102(4), 995–1000.
- Meng, X.-L. (1994). Multiple-imputation inferences with uncongenial sources of input. *Statistical Science*, 9(4), 538–558.
- Mohan, K., & Pearl, J. (2021). Graphical models for processing missing data. *Journal of the American Statistical Association*, 116(534), 1023–1037.
- Molenberghs, G., Beunckens, C., Sotto, C., & Kenward, M. G. (2008). Every missingness not at random model has a missingness at random counterpart with equal fit. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 70(2), 371–388.
- Molenberghs, G., Fitzmaurice, G., Kenward, M. G., Tsiatis, A., & Verbeke, G. (2014). *Handbook of missing data methodology*. CRC Press.
- Moreno-Betancur, M., Lee, K. J., Leacy, F. P., White, I. R., Simpson, J. A., & Carlin, J. B. (2018). Canonical causal diagrams to guide the treatment of missing data in epidemiologic studies. *American Journal of Epidemiology*, 187(12), 2705–2715.
- Morris, T. P., White, I. R., & Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38(11), 2074–2102.

- Neyman, J. (1934). On the two different aspects of the representative method: the method of stratified sampling and the method of purposive selection. *Journal of the Royal Statistical Society*, 97(4), 558–625.
- Nguyen, C. D., Carlin, J. B., & Lee, K. J. (2017). Model checking in multiple imputation: An overview and case study. *Emerging Themes in Epidemiology*, 14(1), 8.
- Nijman, S. W. J., Hoogland, J., Groenhouf, T. K. J., Brandjes, M., Jacobs, J. J. L., Bots, M. L., Asselbergs, F. W., Moons, K. G. M., & Debray, T. P. A. (2020). Real-time imputation of missing predictor values in clinical practice. *European Heart Journal - Digital Health*, 2, 154.
- Oberman, H. I., van Buuren, S., & Vink, G. (2021). Missing the point: Non-convergence in iterative imputation algorithms. *arXiv preprint arXiv:2110.11951*.
- Pawel, S., Kook, L., & Reeve, K. (2022). Pitfalls and potentials in simulation studies. *arXiv preprint arXiv:2203.13076*.
- Petersen, M. L., & Van der Laan, M. J. (2014). Causal models and learning from data: Integrating causal modeling and statistical estimation. *Epidemiology*, 25(3), 418–426.
- Raghunathan, T. E., Reiter, J. P., & Rubin, D. B. (2003). Multiple imputation for statistical disclosure limitation. *Journal of Official Statistics*, 19(1), 1.
- Reiter, J. P. (2003). Inference for partially synthetic, public use microdata sets. *Survey Methodology*, 29(2), 181–188.
- Robins, J. M., & Gill, R. D. (1997). Non-response models for the analysis of non-monotone ignorable missing data. *Statistics in Medicine*, 16(1), 39–56.
- Rubin, D. B. (1976). Inference and missing data. *Biometrika*, 63(3), 581–592.
- Rubin, D. B. (1986). Statistical matching using file concatenation with adjusted weights and multiple imputations. *Journal of Business & Economic Statistics*, 4(1), 87–94.
- Rubin, D. B. (1987). *Multiple imputation for nonresponse in surveys*. Wiley Series in Probability and Mathematical Statistics Applied Probability and Statistics. Wiley.
- Schomaker, M. (2021). Regression and causality. *arXiv preprint arXiv:2006.11754*.
- Schouten, R. M., Lugtig, P., & Vink, G. (2018). Generating missing values for simulation purposes: A multivariate amputation procedure. *Journal of Statistical Computation and Simulation*, 88(15), 2909–2930.
- Schouten, R. M., & Vink, G. (2021). The dance of the mechanisms: How observed information influences the validity of missingness assumptions. *Sociological Methods & Research*, 50(3), 1243–1258.
- Seaman, S., Galati, J., Jackson, D., & Carlin, J. (2013). What is meant by “missing at random”? *Statistical Science*, 28(2), 257–268.
- Shara, N., Yassin, S. A., Valaitis, E., Wang, H., Howard, B. V., Wang, W., Lee, E. T., & Umans, J. G. (2015). Randomly and non-randomly missing renal function data in the strong heart study: A comparison of imputation methods. *PLoS One*, 10(9), e0138923.
- Sperrin, M., Martin, G. P., Sisk, R., & Peek, N. (2020). Missing data should be handled differently for prediction than for description or causal explanation. *Journal of Clinical Epidemiology*, 125, 183–187.
- Van Buuren, S. (2018). *Flexible imputation of missing data*. Chapman and Hall/CRC.
- Van Buuren, S., & Groothuis-Oudshoorn, K. (2011). Mice: Multivariate imputation by chained equations in R. *Journal of Statistical Software*, 45(1), 1–67.
- Vink, G. (2022). Strategies for simulating missingness. <https://www.gerkovink.com/simulate>. *GitHub repository*. <https://doi.org/10.5281/zenodo.7467995>
- Vink, G., & Van Buuren, S. (2014). Pooling multiple imputations when the sample happens to be the population. *arXiv preprint arXiv:1409.8542*.
- Volker, T. B., & Vink, G. (2021). Anonymized shareable data: Using mice to create and analyze multiply imputed synthetic datasets. *Psych*, 3(4), 703–716.
- Wood, A. M., Royston, P., & White, I. R. (2015). The estimation and use of predictions for the assessment of model performance using large samples with multiply imputed data. *Biometrical Journal Biometrische Zeitschrift*, 57(4), 614–632.
- Zhao, Y. (2022). Diagnostic checking of multiple imputation models. *ASIA advances in statistical analysis*, 106(2), 271–286.
- Zhu, J., & Raghunathan, T. E. (2015). Convergence properties of a sequential regression multiple imputation algorithm. *Journal of the American Statistical Association*, 110(511), 1112–1124.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Oberman, H. I., & Vink, G. (2024). Toward a standardized evaluation of imputation methodology. *Biometrical Journal*, 66, 2200107. <https://doi.org/10.1002/bimj.202200107>

APPENDIX

A.1 | Reporting guidelines

Table A1 provides a checklist for reporting on imputation methodology evaluations. We have structured the checklist according to simulation study reporting format proposed by Morris et al. (2019).

TABLE A1 Checklist for reporting on imputation methodology evaluations.

☐	Aims
	e.g., simulation scope
	☐ simulation design (incl. pseudocode or flow diagram)
	☐ required level of precision (incl. number of simulation repetitions)
☐	Data-generating mechanisms
	e.g., data generation and missingness generation
	☐ data source (incl. model-based or design-based, sampling variance)
	☐ data characteristics (incl. multivariate relations and data structures such as clustering)
	☐ missingness mechanisms (incl. type or functional form of the missing data model)
	☐ missingness patterns (incl. missingness proportion)
☐	Estimands
	e.g., complete data target
☐	Methods
	e.g., missing data methods, analytic method and inference pooling rules
	☐ imputation methods (incl. parameters such as the number of imputations)
	☐ estimation method (incl. reference methods such as complete case analysis)
	☐ methods used to construct standard errors and confidence intervals
	– Such as rules by Barnard and Rubin (1999); Rubin (1987); Reiter (2003), etc.
☐	Performance measures
	e.g., evaluation of estimates and imputed values
	☐ statistical properties (incl. comparative performance, if applicable)
	☐ validity of imputations (incl. imputation model fit and distributional characteristics)
	– Does the imputation model fit well to the observed values? (e.g., posterior predictive check)
	– Are there unrealistic values in the data? (e.g., negative where positive expected)
	– Are there implausible combinations in the data? (e.g., pregnant grandfathers)

3 Chapter 3: convergence

[TODO: link/insert + credit]

test

4 Chapter 4: ggmmice

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test

5 Discussion

In this dissertation, I have investigated how to evaluate imputation methodology. Instead of seeing imputation as a quick fix for treating missing data, I have approached it as a sociotechnical system: a methodological practice that requires continuous assessment and reflection. A central theme that runs through all chapters is that imputation methodology should be evaluated, whether computationally or graphically. Evaluation is key both for methodologists who develop imputation methodology, as well as for imputers who apply imputation in practice.

5.1 Main findings

The central theme of this dissertation is that imputation methodology requires verification and validation. Verification happens in simulation studies by means of computational evaluation of imputation methods. Validation happens in practice, by means of computational evaluation and graphical evaluation of the imputation workflows.

In chapter 2, we learned that testing new imputation methods through simulation studies requires careful, standardized design. If this is not the case, simulation studies might not cover all relevant conditions, and results are not comparable between studies. To make imputation methods comparable across studies, we must standardize simulation designs so that different methods face similar, clearly documented scenarios, report design choices transparently, and publish research compendiums so that others can reproduce and extend the experiments. Even with such care, there is a limit to what simulation studies can tell us. We can know that a method worked under the scenarios we simulated, but we cannot be certain that it will work in practice. That is why we need validation.

In chapter 3 we studied the algorithmic convergence of the `mice` algorithm. Diagnosing non-convergence quantitatively is a form of computational evaluation of the imputation workflow. Since `mice` converges in distribution, and not to a point, non-convergence is hard to diagnose. We learned that commonly used metrics and thresholds for non-convergence in iterative algorithms such as `mice`, may not be directly applicable to FCS. Simulation results show that MICE reaches stable output before non-convergence metrics would indicate so. There is a mismatch between theoretical convergence criteria and practical algorithmic behavior. That is why we still have to rely on the current standard advised evaluation tool for algorithmic convergence: visual inspection. Inspecting traceplots for signs of non-convergence is a form of visual diagnostic evaluation of imputation workflows, or graphical evaluation. Graphical evaluation was the topic of the fourth chapter.

Visual inspection aids imputers in developing and evaluating imputation models, because formal tests are not available. The fifth chapter introduces the R package `ggmice`. `ggmice` offers tools for the graphical evaluation of the imputation models that were previously unavailable. `ggmice` functions aid imputers in visualizing incomplete data and imputed data side-by-side,

displaying missingness and imputation models in an interpretable way, and by producing plots that can be extended and customized for publication. These visual diagnostics support existing evaluations whenever quantitative metrics and substantive reasoning are unfeasible.

5.2 Recommendations

For colleagues who design and study imputation methods, this dissertation suggests the following:

- Design simulation studies in a structured and reproducible manner. Make sure that the simulation conditions and performance measures cover all relevant scenarios and are comparable across methods. Report all design choices, and publish code (and data if applicable) alongside methodological papers.
- Engage with applied researchers. Talk to your intended audience (i.e., applied researchers), or at least find out what they are struggling with via discussion forums or surveys. This should inform the design of methods and diagnostics.

For researchers who implement imputation in their own studies, the main advice can be summarized as:

- Think before you code. Do not treat `mice` or any other package as a black box that magically treats your missing data problem. Consider the missingness mechanisms and your analysis model(s) before specifying imputation methods and predictor sets. Your substantive knowledge is valuable for treating the missingness.
- Look at the data, plot the data. Inspect incomplete variables, missingness patterns, and imputed values using graphics. Use visualization to inform model building, diagnose misfit, and communicate uncertainty.
- Treat diagnostics as part of the analysis, not an afterthought. Convergence plots, distributional comparisons, and sensitivity analyses should be integrated into the workflow and reported.
- Please cite the software you use. For R packages, it's as easy as running `citation("ggmice")`.

5.3 Limitations

Computational and graphical evaluation are not a substitute for thinking. Simulation studies, diagnostic metrics and plots can offer detailed insight into imputation workflows, but they may also create a false sense of certainty. For example, imputed values that lie within the range of the observed data may give a false sense of certainty against a MNAR mechanism, which is an untestable assumption. No amount of computation or visualization can prove that the missing-data mechanism is correctly assumed. Instead, one should use computational evaluation as complement to (not a replacement for) substantive knowledge and expert insight.

This dissertation investigates imputation workflows at the methodological level, without systematic user feedback. I have not tested the evaluation tools I propose experimentally. Assessing their causal impact on user behavior and imputation quality in practice was not feasible. I did receive feedback via teaching, workshops, collaborations, and online forums GitHub and StackOverflow. This informal evaluation is an indication of user appreciation(?). And I have used this input for example to further develop `ggmice`. So even though it might be impossible to draw evidence-based conclusions about the efficacy of the tools I developed, we may still call the development evidence-informed, maybe?

Future direction: automated sensitivity analyses and over-imputation. Sensitivity analyses that vary model assumptions, include alternative predictors, or impose plausible MNAR structures can reveal how fragile conclusions are to changes in imputation choices. Over-imputation can reveal imputation model misfit on observed part of the data. Since publishing, there has not been a pre-reg format

Since this dissertation relies on the R language and centers the `mice` package, it is not certain how results translate to the wider imputation ecosystem. Concepts may generalize beyond `mice` in R. The focus reflects the current dominance of R and `mice` in many applied fields, and allows for depth and specificity in tool development. While this set-up is popular, the data science landscape is expanding, with many machine learning and deep learning methods emerging as candidate imputation engines. These methods are not (yet) widely implemented within `mice`. They might be able to better approximate the posterior predictive distribution of the missing values than `mice` methods, but this requires dedicated research and evaluation, which is outside of the scope of this dissertation.

Some other directions for future research include computational evaluation of variable importance within imputation models, integration of `ggmice` with `naniar`, and better support for non-numeric variables. The visualizations implemented in `ggmice` are generally better suited for numeric data, as opposed to categorical variables (or even open text field data, which is not supported by `mice` currently). Future research may focus on associations of categorical variables with missingness indicators, and convergence parameter other than SD of imputed values.

5.4 Outlook

Missingness will be a persistent problem in research data. There is no foreseeable future in which all empirical datasets are complete. I therefore expect imputation methodology to continue the way we deal with data might change. But the way we deal with data might change.

With the rise of machine learning and deep learning methods (or “AI”), evaluation will become ever more important. These new methods will also require new verification and validation tools, especially if the framework for intended applications changes. Many novel methods are designed under a prediction framework, not a statistical inference framework. Instead of inferring from a sample to the population, the aim becomes to infer about an

individual case in the future. Methods intended for prediction may excel at point prediction, but ignore uncertainty in estimates. Omitting sampling variance from the estimates and to failing to incorporate between-imputation variance underrepresents the uncertainty. When such methods are used naively for inference, they produce confidence intervals that are too narrow and p -values that are too small. The result is an increased risk of spurious findings. Therefore, all models must propagate uncertainty.

Any model used in scientific inference should propagate uncertainty. This includes (but is not limited to) integrating missing-data uncertainty into model outputs and reporting interval estimates that align with the uncertainty in the data and the modeling process. This also holds for graphical presentations.

Trust in science requires an honest reflection of uncertainty in our analyses. So please be open about uncertainty and limitations of scientific studies. And while we're at it, also please publish null results (e.g., via registered reports or pre-registrations), share data and code with other scientists and the public, and publish open access (including educational materials). At the same time, we should change the way we evaluate science. This dissertation is an example of the new 'recognition and rewards' vision. It includes software as an actual chapter, not something extra. This is part of the current movement of valuing diverse forms of contribution beyond traditional papers. I hope this can serve as an example for other PhD candidates and researchers to reflect on what they consider scientific output. I hope and to claim space for software, open data, and methodological infrastructure as central achievements. I hope to inspire others to claim space for 'alternative' output such as software and data as full-fledged research output.

Finally, it is important to note that imputation is not always the right response to missingness. Missing data can signal a mismatch between our data collection methods and the lived experiences of participants. Systematic patterns of non-response may point to issues in the survey design, accessibility, or trust. In such cases, treating missingness solely as a statistical nuisance to be 'fixed' risks overlooking important social and ethical dimensions of research, especially in the social and biomedical sciences, where rows in the data represent human beings. I would love to study missingness as a method to tell you about mismatches between data collection and participant perspectives.

In short, I hope this dissertation encourages to question defaults, to make evaluation practices explicit, and to contribute to a scientific culture that takes uncertainty—and our responsibilities in representing it—seriously.

References

- Allison, P. D. (2002). *Missing Data*. SAGE Publications, Inc. <https://doi.org/10.4135/9781412985079>
- Alsalti, T., Rohrer, J. M., & Arslan, R. C. (2026). Thinking Clearly About Sampling and Representation With the Total Survey Error Framework. *Social and Personality Psychology Compass*, 20(7), e70160. <https://doi.org/10.1111/spc3.70160>
- Bartlett, J. W., Seaman, S. R., White, I. R., & Carpenter, J. R. (2015). Multiple imputation of covariates by fully conditional specification: Accommodating the substantive model. *Statistical Methods in Medical Research*, 24(4), 462–487. <https://doi.org/10.1177/0962280214521348>
- Carpenter, J. R., & Smuk, M. (2021). Missing data: A statistical framework for practice. *Biometrical Journal*, 63(5), 915–947. <https://doi.org/10.1002/bimj.202000196>
- Curzon, P., Dorling, M., Ng, T., Selby, C., & Woollard, J. (2014). *Developing computational thinking in the classroom: A framework*. Computing at School.
- de Ruiter, J. P., & Albert, S. (2017). An Appeal for a Methodological Fusion of Conversation Analysis and Experimental Psychology. *Research on Language and Social Interaction*, 50(1), 90–107. <https://doi.org/10.1080/08351813.2017.1262050>
- Deming, W. E. (1944). On Errors in Surveys. *American Sociological Review*, 9(4), 359–369. <https://doi.org/10.2307/2085979>
- Dudek, G. (2004). Computational Evaluation. In G. Dudek (Ed.), *Collaborative Planning in Supply Chains: A Negotiation-Based Approach* (pp. 165–213). Springer. https://doi.org/10.1007/978-3-662-05443-7_7
- Evaluation measures (information retrieval). (2025). In *Wikipedia*. [https://en.wikipedia.org/w/index.php?title=Evaluation_measures_\(information_retrieval\)&oldid=1329650960](https://en.wikipedia.org/w/index.php?title=Evaluation_measures_(information_retrieval)&oldid=1329650960)
- Federal Committee on Statistical Methodology. (2001). *Statistical Policy Working Paper 31. Measuring and Reporting Sources of Error in Surveys*. <https://doi.org/10.21949/1529874>
- Gelman, A., & Shalizi, C. R. (2013). Philosophy and the practice of Bayesian statistics. *British Journal of Mathematical and Statistical Psychology*, 66(1), 8–38. <https://doi.org/10.1111/j.2044-8317.2011.02037.x>
- Graham, J. W. (2012). *Missing Data*. Springer. <https://doi.org/10.1007/978-1-4614-4018-5>
- Hand, D. (2020). *Dark data: Why what you don't know matters*. Princeton University Press.
- Hand, D. (2025). Statistics: An Overview. In *International Encyclopedia of Statistical Science* (pp. 2654–2659). Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-69359-9_649
- IEEE Standard for System, Software, and Hardware Verification and Validation (2025). <https://doi.org/10.1109/IEEESTD.2025.11134780>
- ISO, & IEC. (2015). *ISO/IEC 2382:2015 — information technology — vocabulary*. ISO; IEC; International Standard;
- Little, R. J. A., & Rubin, D. B. (2019). *Statistical analysis with missing data. Third edition*. John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781119482260>
- Little, & Rubin. (2020). *Statistical Analysis with Missing Data, Third Edition | Wiley Series in Probability and Statistics*. <https://onlinelibrary.wiley.com/doi/book/10.1002/>

9781119482260

- Liu, D., Oberman, H. I., Muñoz, J., Hoogland, J., & Debray, T. P. A. (2023). Quality Control, Data Cleaning, Imputation. In F. W. Asselbergs, S. Denaxas, D. L. Oberski, & J. H. Moore (Eds.), *Clinical Applications of Artificial Intelligence in Real-World Data* (pp. 7–36). Springer International Publishing. https://doi.org/10.1007/978-3-031-36678-9_2
- Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511809071>
- Molenberghs, G., Fitzmaurice, G., Kenward, M. G., Tsiatis, A., & Verbeke, G. (Eds.). (2014). *Handbook of Missing Data Methodology*. Chapman and Hall/CRC. <https://doi.org/10.1201/b17622>
- Neyman, J. (1934). On the Two Different Aspects of the Representative Method: The Method of Stratified Sampling and the Method of Purposive Selection. *Journal of the Royal Statistical Society*, 97(4), 558–625. <https://doi.org/10.2307/2342192>
- Nguyen, C. D., Carlin, J. B., & Lee, K. J. (2017). Model checking in multiple imputation: An overview and case study. *Emerging Themes in Epidemiology*, 14(1), 8. <https://doi.org/10.1186/s12982-017-0062-6>
- Nguyen, C. D., Carlin, J. B., & Lee, K. J. (2021). Practical strategies for handling breakdown of multiple imputation procedures. *Emerging Themes in Epidemiology*, 18, 5. <https://doi.org/10.1186/s12982-021-00095-3>
- Oberman, H. I. (2022). *Ggmice: Visualizations for 'mice' with 'Ggplot2'* [Computer software]. Zenodo. <https://doi.org/10.5281/zenodo.6532702>
- Oberman, H. I., University, U., Utrecht, U. M. C., Volker, T., Vink, G., Vink, P., Wallis, J., & Lang, K. (2025). *Ggmice: Visualizations for 'mice' with 'Ggplot2'* (Version 0.1.1) [Computer software]. <https://cran.r-project.org/web/packages/ggmice/index.html>
- Rosenberg, A. (2018). *Philosophy of Social Science*. Routledge. <https://books.google.com?id=GRdWDwAAQBAJ>
- Rubin, D. B. (1976). Inference and Missing Data. *Biometrika*, 63(3), 581–592. <https://doi.org/10.2307/2335739>
- Rubin, D. B. (1987). *Multiple Imputation for nonresponse in surveys*. Wiley.
- Rubin, D. B. (1996). Multiple Imputation after 18+ Years. *Journal of the American Statistical Association*, 91(434), 473–489. <https://doi.org/10.1080/01621459.1996.10476908>
- Schafer, J. L. (1997). *Analysis of Incomplete Multivariate Data*. Chapman and Hall/CRC. <https://doi.org/10.1201/9781439821862>
- Schafer, J. L., & Graham, J. W. (2002). Missing data: Our view of the state of the art. *Psychological Methods*, 7(2), 147–177. <https://www.ncbi.nlm.nih.gov/pubmed/12090408>
- Sommerville, I. (2015). *Software engineering* (Tenth edition). Pearson.
- van Buuren, S. (2007). Multiple imputation of discrete and continuous data by fully conditional specification. *Statistical Methods in Medical Research*, 16(3), 219–242. <https://doi.org/10.1177/0962280206074463>
- van Buuren, S. (2018). *Flexible imputation of missing data* (2nd ed.). CRC Press.
- van Buuren, S., Brand, J. P. L., Groothuis-Oudshoorn, C. G. M., & Rubin, D. B. (2006). Fully conditional specification in multivariate imputation. *Journal of Statistical Computation and Simulation*, 76(12), 1049–1064. <https://doi.org/10.1080/10629360600810434>

- van Buuren, S., & Groothuis-Oudshoorn, C. G. M. (1999). *Flexible multivariate imputation by MICE* (Technical Report No. PG/VGZ/99.054). TNO Prevention and Health.
- van Buuren, S., & Groothuis-Oudshoorn, K. (2011). mice: Multivariate Imputation by Chained Equations in R. *Journal of Statistical Software*, 45(3), 1–67. <https://doi.org/10.18637/jss.v045.i03>
- Wilkinson, L. (2012). The Grammar of Graphics. In *Handbook of Computational Statistics* (pp. 375–414). Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-21551-3_13

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